

Name: _____

Double Award Science – Physics Unit 1 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	G	F	E	D	C	Max UMS
2017	7	10	13	17	21	25
2018	5	9	13	17	22	27
2019	10	14	18	22	26	30
2022	6	11	16	21	27	33
2023	13	17	21	26	31	36

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3430U30-1



SCIENCE (Double Award)

Unit 3: PHYSICS 1
FOUNDATION TIER

MONDAY, 19 JUNE 2017 – MORNING
1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	8	
3.	12	
4.	8	
5.	9	
6.	15	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper, you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the continuation page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 5(a) is a quality of extended response (QER) question where your writing skills will be assessed.



JUN173430U30101

Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	

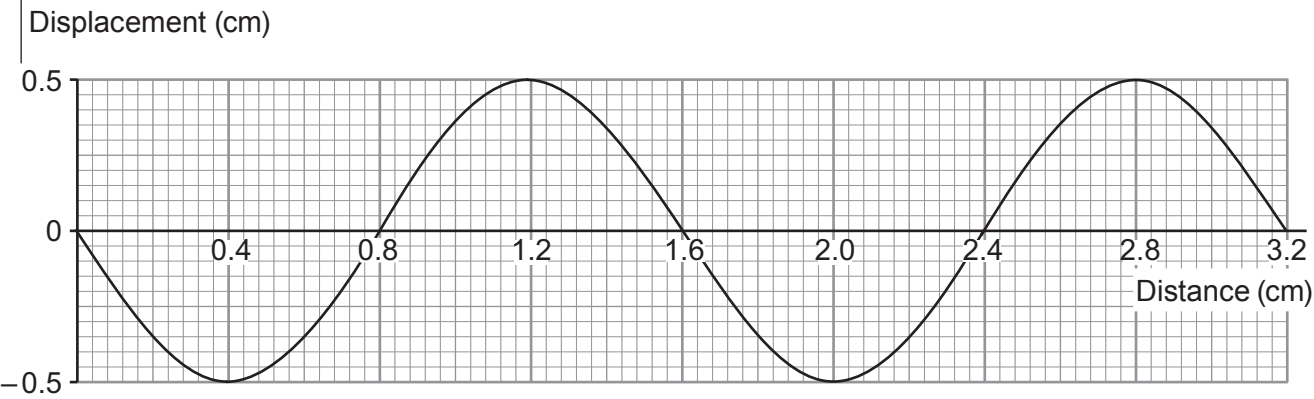
SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6



Answer all questions.

1. A teacher demonstrates water waves in a ripple tank. The diagram represents the waves.

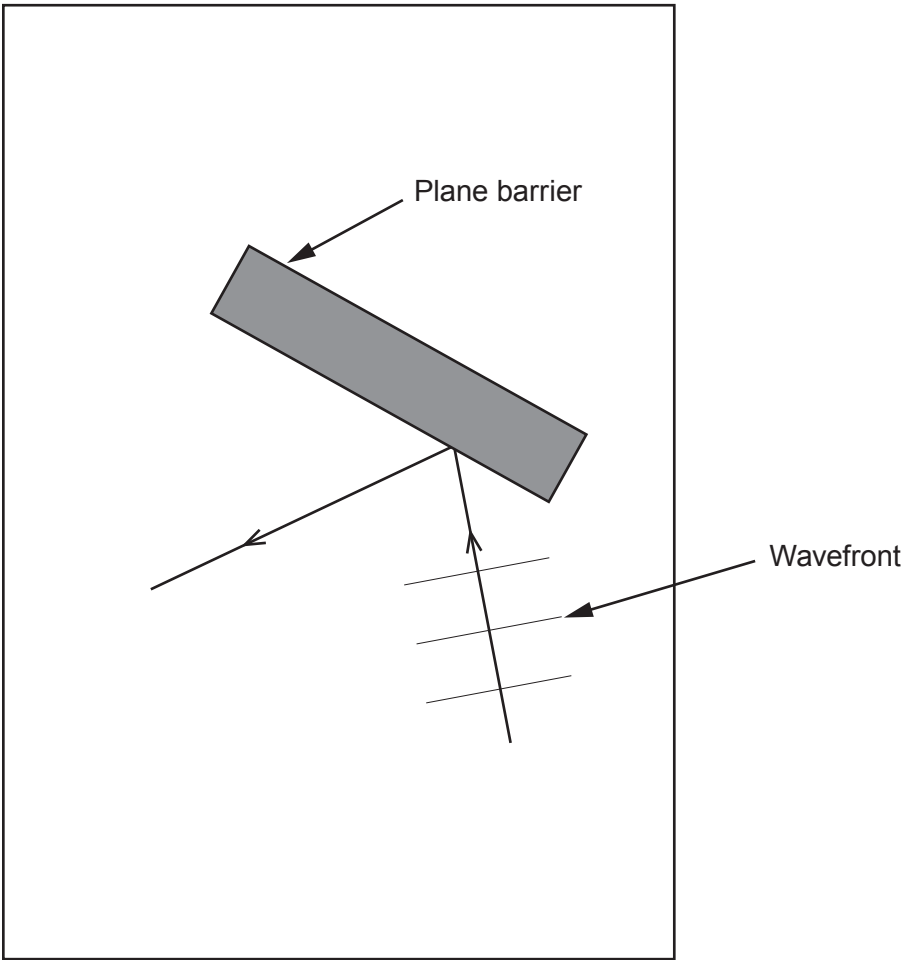


- (a) (i) State the wavelength of the wave. cm [1]
- (ii) State the amplitude of the wave. cm [1]
- (b) It is observed that 10 waves pass a point in the tank in 2 s.
- (i) Underline the correct value for the frequency of the waves.
- 2 Hz 5 Hz 10 Hz 20 Hz [1]
- (ii) Select an equation from page 2 and use it to calculate the wave speed. [3]

Wave speed = cm/s



(c) The waves are made to reflect off a plane barrier. **Complete the diagram** to show the reflected wavefronts. [2]



8



Examiner
only

2. Electromagnetic (em) waves are widely used in a variety of ways. Satellites rely on microwaves to communicate with Earth as the microwaves penetrate through the upper layers of our atmosphere. Infra-red waves have a higher frequency than microwaves and are used widely in communications. X-rays and gamma rays are ionising radiation and are both used in medicine; gamma rays can be used in cancer treatment, shrinking tumours. Radio waves are used to transmit TV and radio signals.

(a) The diagram shows an incomplete em spectrum. **Complete the diagram** to show the em spectrum in order. [2]

Radio waves			Visible light	UV light		Gamma rays
-------------	--	--	---------------	----------	--	------------

(b) All em waves are transverse. Describe what is meant by a transverse wave. [2]

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(c) Tick (✓) the boxes alongside the **two** correct statements about em waves below. [2]

- Only radio waves and microwaves transmit information☐
- All em waves travel at the same speed in a vacuum☐
- All em waves are a form of radiation☐
- All em waves have the same frequency☐
- Radio waves have the smallest wavelength☐

(d) A student suggests that radio waves are just as harmful to humans as gamma rays. Explain whether or not you agree with this statement. [2]

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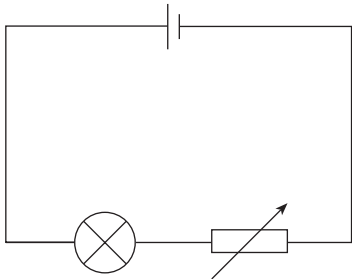
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3. (a) Students are investigating the current-voltage characteristic of a filament lamp. An incomplete diagram of their circuit is shown below.



- (i) **Add to the diagram** a correctly connected ammeter and voltmeter. [2]
- (ii) Describe how the circuit is used to obtain a series of measurements of current and voltage. [3]

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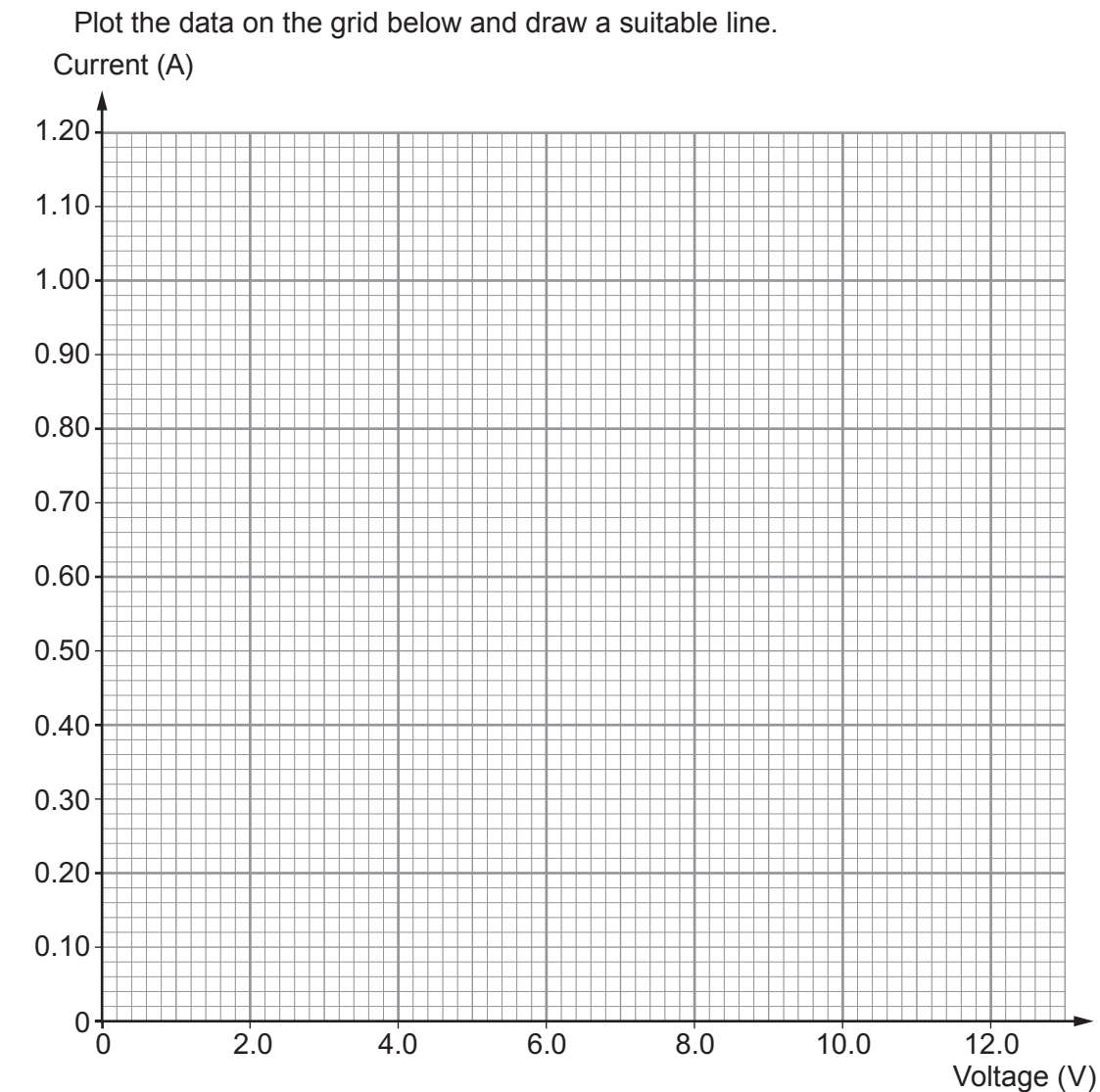
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- (b) The following data were collected using the circuit.

Voltage (V)	Current (A)
0.0	0.00
2.0	0.46
4.0	0.76
6.0	0.94
8.0	1.02
10.0	1.06
12.0	1.08



Examiner
only



(c) (i) Use the graph and the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the lamp when the voltage across it is 3 V. [2]

Resistance = Ω

(ii) The students conclude that the **resistance** of the lamp increases as voltage increases. Explain how the data support this conclusion. [2]

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4. The UK relies on a continuous supply of electricity which is provided by a variety of different power stations. Electricity can be generated in power stations and transmitted to users through the National Grid.

One concern with electricity production is the amount of greenhouse gases produced. Increased levels of these gases in our atmosphere is leading to climate change. Renewable energy resources produce less greenhouse gases but are not able to provide a constant supply of electricity.

(a) (i) **Draw a line** from each method of electricity generation on the left to the most likely disadvantage on the right. [2]

Method of generating electricity

Nuclear power station
Wind farm
Coal-fired power station

Disadvantage

High CO ₂ emissions
High decommissioning cost
Unreliable

(ii) The energy input to a nuclear power station is 20 MJ and its efficiency is 40%. The energy input to a coal-fired power station is 20 MJ and its useful output energy is 8 MJ.
A student suggests that the nuclear power station is more efficient than the coal-fired power station as it wastes less energy. Explain whether or not you agree with this conclusion. (You may wish to use an equation from page 2.) [2]

Space for calculations

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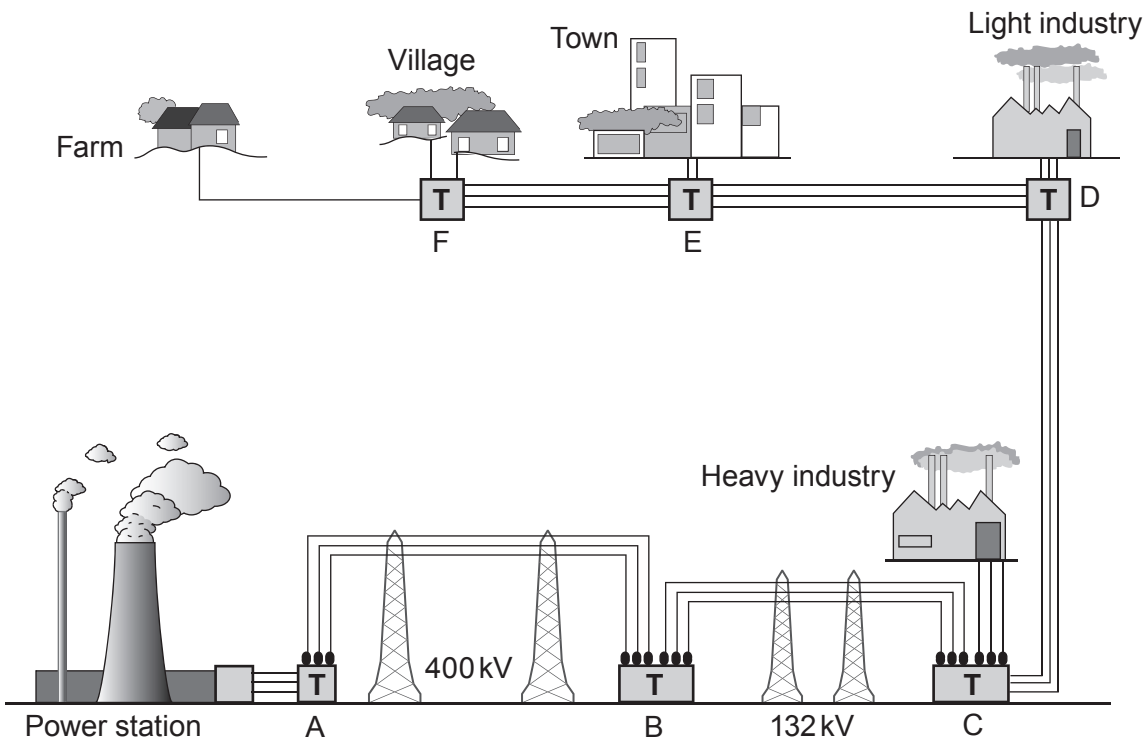
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(b) The diagram shows a part of the National Grid.

Key

T – transformer



(i) State **one** advantage of the National Grid. [1]

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(ii) Which of the transformers A, B, C, D, E or F is a step-up transformer? [1]

(iii) Explain why step-up transformers are used in the National Grid. [2]

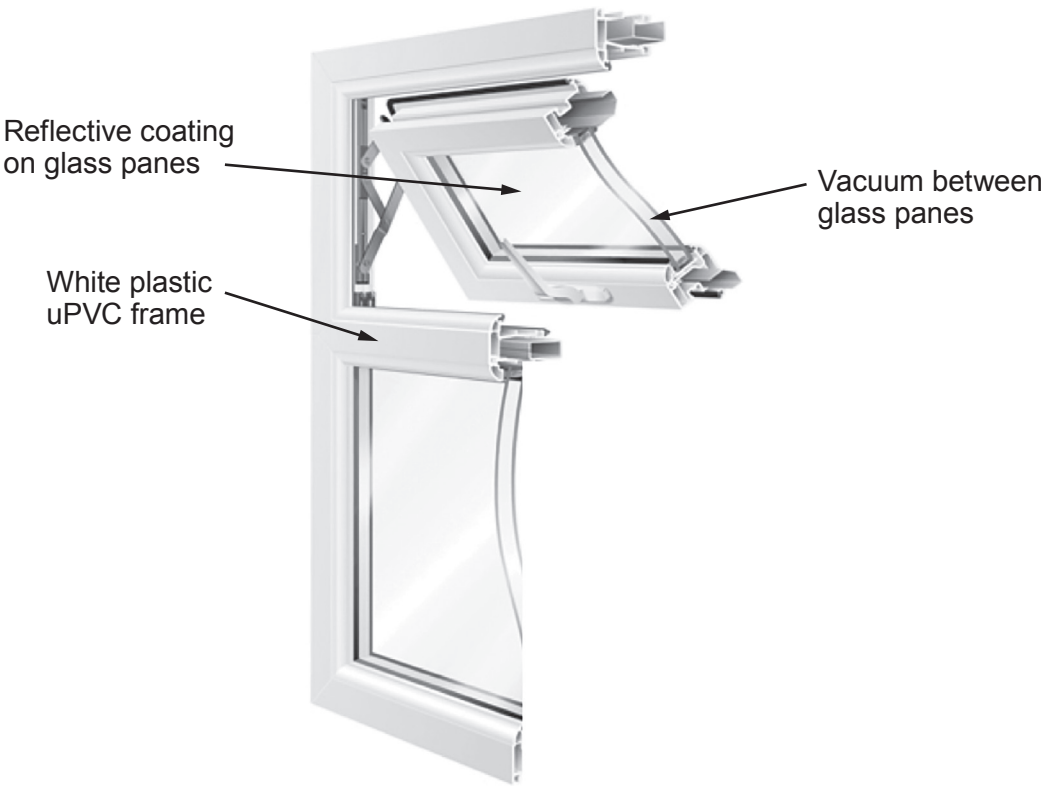
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5. (a) A householder decides to replace his single glazed, aluminium framed windows. A company recommends fitting 'Energy Windows' like the one shown below.



Explain in terms of conduction, convection and radiation how the labelled features will help to reduce energy losses from the house. [6 QER]

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Examiner
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(b) A family fits double glazing to their house at a cost of £4 000. They save £80 per year on their heating bill.

(i) Calculate the payback time. [1]

Payback time = years

(ii) Suggest **two** reasons why this payback time may change. [2]

1.

2.

9



6. The Energy Saving Trust aims to help everyone save energy every day in order to reduce the environmental damage that energy production does to the environment. It provides advice to households and businesses and gives tips for energy savings such as not leaving appliances on standby and choosing the most energy efficient appliances.

The efficiency of washing machines depends on how they are used as they can wash clothes at a variety of temperatures. Many clothes retailers and washing detergent manufacturers recommend washing clothes at 30°C in order to be more energy efficient. According to the Energy Saving Trust “*Washing clothes at 30°C rather than at higher temperatures uses around 40% less energy.*”

The table below gives data for different washing temperatures for a particular washing machine.

Washing temperature (°C)	Mean power (W)	Wash time (minutes)
30	500	60
40	500	90
50	500	120

(a) (i) Use equations from page 2 to calculate the cost of washing clothes at 40°C. The cost of one unit of electricity is 16 p. [4]

Cost = p

(ii) The Energy Saving Trust suggests that washing clothes at 30°C reduces energy use by at least 40% compared to washing at higher temperatures. Use the data in the table to investigate the validity of this claim. [3]

Space for calculations

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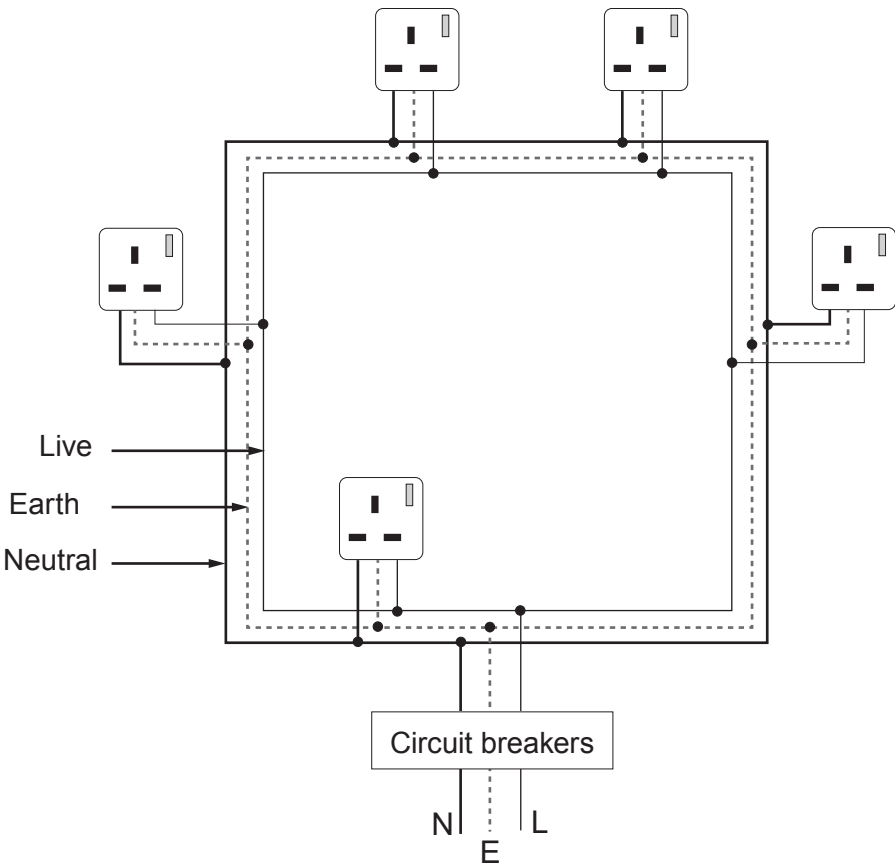
(iii) Washing clothes at 30°C saves money on household bills. Explain why reducing energy consumption also helps the environment. [2]

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(b) The washing machine is connected to the ring main shown below. The voltage in the ring main is 230 V.



(i) State which wire the fuse must be connected to. [1]

.....

(ii) Describe the function of the earth wire. [2]

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TURN OVER FOR THE LAST PART OF THE QUESTION.



- (c) When the voltage is 230 V, the mean power of the washing machine during a wash is 500 W. However when heating the water it must be able to produce a maximum power of 2400 W. The householder has 3 A, 5 A and 13 A fuses available and chooses to fit a 3 A fuse into the plug. Investigate whether or not this is a suitable choice. [3]

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END OF PAPER

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15



Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3430U30-1



SCIENCE (Double Award)

Unit 3 – PHYSICS 1
FOUNDATION TIER

FRIDAY, 15 JUNE 2018 – MORNING

1 hour 15 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	4	
2.	6	
3.	12	
4.	11	
5.	12	
6.	15	
Total	60	

ADDITIONAL MATERIALS

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INSTRUCTIONS TO CANDIDATES

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INFORMATION FOR CANDIDATES

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The assessment of the quality of extended response (QER) will take place in question **3(a)**.



JUN183430U30101

Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	

SI multipliers

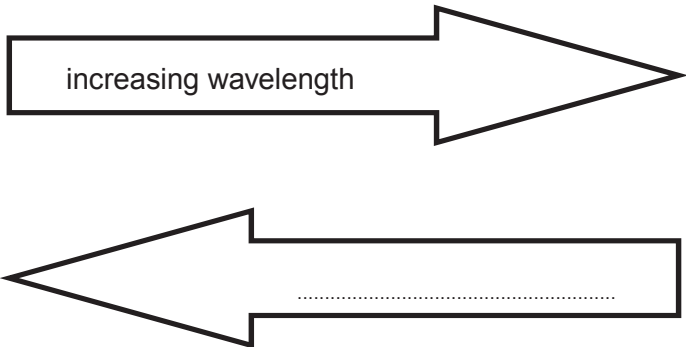
Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6



Answer all questions.

1. Various types of radiation form a continuous spectrum called the electromagnetic (em) spectrum. The diagram shows the regions of the em spectrum.

Gamma rays	X-rays	Ultraviolet	Visible light	A	Microwaves	Radio waves
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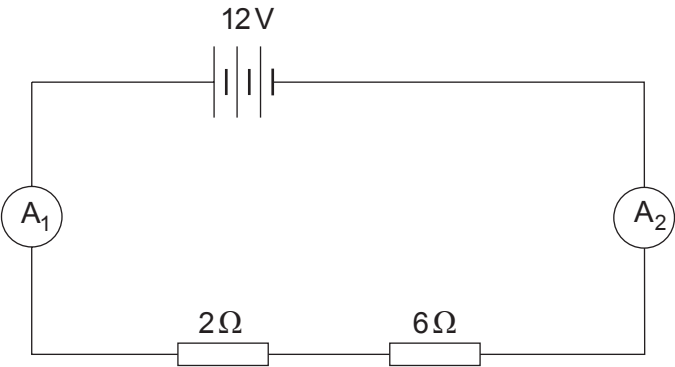


- (a) (i) Complete the arrow above, using **only one** of the following phrases. [1]
- decreasing speed
 - increasing speed
 - increasing frequency
 - decreasing energy
- (ii) Name the em radiation labelled **A** in the diagram. [1]
-
- (b) (i) Name the ionising em wave given out by some radioactive materials. [1]
-
- (ii) State why ionising em radiation is dangerous to humans. [1]
-

3430U301
03



2. In class, a group of students set up a simple series circuit as shown in the diagram below.



(a) (i) Use the equation:

$$R = R_1 + R_2$$

to calculate the total resistance in the circuit. [1]

Total resistance = Ω

(ii) Use an equation from page 2 to calculate the current reading on ammeter A₁. [2]

Ammeter A₁ reading = A

(iii) State the current reading on ammeter A₂. [1]

Ammeter A₂ reading = A

(b) The 6Ω resistor is now removed from the circuit. It is added **in parallel** with the 2Ω resistor. The circuit is complete.

Complete the following sentences by underlining the correct phrase or word. [2]

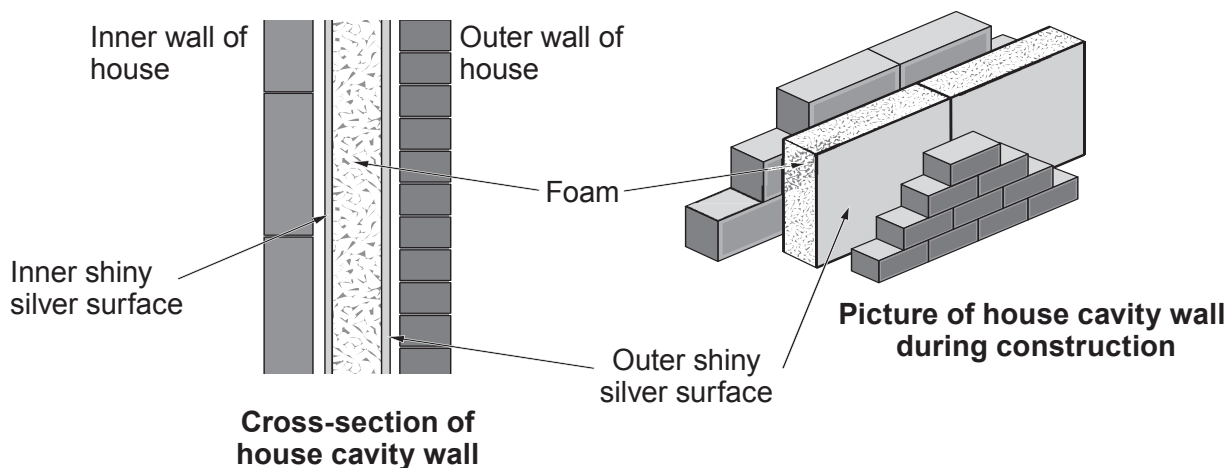
- (i) The total resistance of the circuit (**increases** / **stays the same** / **decreases**).
- (ii) The current reading on the ammeter A₁ (**increases** / **stays the same** / **decreases**).



Examiner
only

- 3.** About 30 % of all the heat lost from an uninsulated house escapes through the walls.

Houses built in Wales since 1990 have wall insulation installed. One type of insulation that is commonly used is cavity board. This is a sheet of solid foam that is covered on both sides with shiny silver foil. During construction, sheets of cavity board are placed between the inner and outer walls of the house.



- (a) Explain why heat is lost through the walls of a house **and** how cavity board can reduce this heat loss by **conduction, convection and radiation**. [6 QER]



(b) The table shows some information about three types of house with cavity board installed. All three houses use gas central heating.

Type of house	Cost of installing cavity board (£)	Saving per year (£)	Carbon footprint without cavity board installed (kg of CO ₂ /year)	CO ₂ saving with cavity board installed (kg/year)	Carbon footprint with cavity board installed (kg of CO ₂ /year)
detached	720	275	5 500	1 100	4 400
semi-detached	480	160	3 640		3 000
mid-terraced	385	110	2 440	440	

(i) The payback time of installing cavity board in a semi-detached house is 3 years. A semi-detached house owner claims that she has a longer payback time than the owner of the **mid-terraced** house. Explain whether you agree with her statement. *Space for workings.* [2]

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(ii) **Complete the table** for the two missing values. *Space for workings.* [2]

(iii) Use information in the table to explain which of the three types of house has the **smallest** impact on the environment. [2]

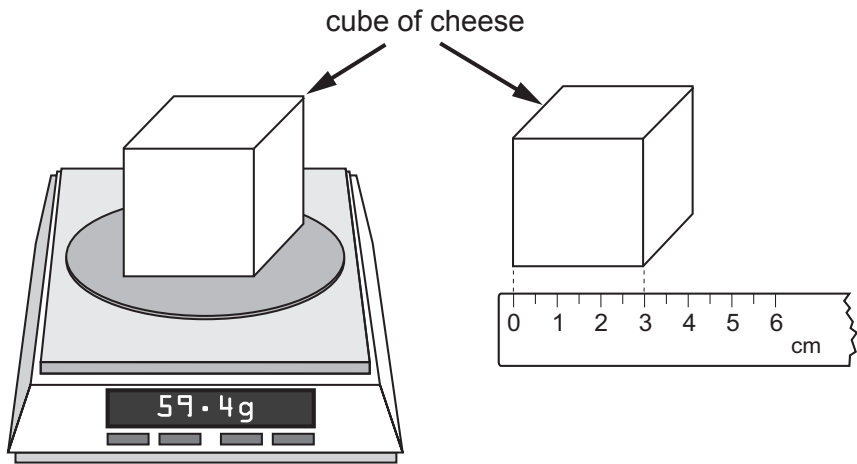
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4. A North Wales cheese producer checks the quality of their cheese by measuring its density. This can be carried out experimentally. The two diagrams show measurements that were carried out on a cube of cheese. The balance read 0.0 g before the cube of cheese was placed on it.



(a) (i) Write down the mass of the cheese cube. g [1]

(ii) Write down the length of one side of the cheese cube. cm [1]

(iii) Use the equation:

volume of a cuboid = length $\times$ width $\times$ height

to calculate the volume of the cheese cube in cm³. [1]

Volume = cm³

(iv) Use the equation:

density = $\frac{\text{mass}}{\text{volume}}$

to calculate the density of the cheese cube and state the unit. [3]

Density = Unit =



- (b) (i) A worker at the cheese factory suggests that the mass of the cheese cube should be measured more than once. State **two** reasons why this is good scientific practice. [2]

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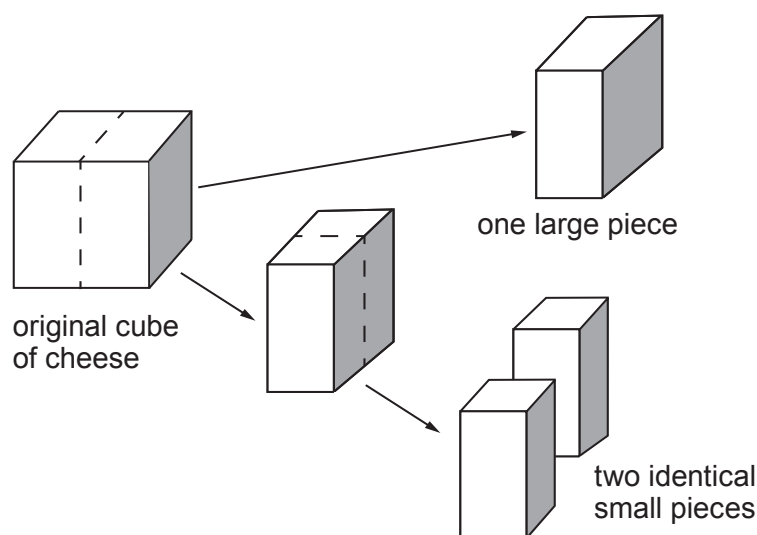
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- (ii) State how the **measurements** could be improved. [1]

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- (c) The cube of cheese used in the experiment is now cut into two identical halves. One of these pieces is then cut in half. The sample of cheese has been divided into three pieces, one large piece and two identical small pieces.



Tick (✓) the **two** correct statements.

[2]

One of the small pieces of cheese is $\frac{1}{3}$ of the mass of the large piece.

☐

One of the small pieces of cheese is $\frac{1}{3}$ of the volume of the large piece.

☐

The three pieces of cheese have the same density.

☐

The mass of the large piece of cheese is half the mass of the original cube.

☐

The mass to volume ratio for each of the three samples is different.

☐


5. The diagram shows an electric hand drier connected to a 230 V a.c. mains supply.



(a) Complete the following sentence by underlining the correct phrase. [1]

An a.c. or alternating current (**continuously changes direction / continuously changes frequency / continuously changes speed**).

(b) The electric hand drier has a label attached to it.

Electric hand drier	
Power = 2600 W	230 V a.c

(i) Each time the hand drier is used it blows out hot air for 15 seconds. Use an equation from page 2 to calculate the energy transferred by the hand drier if it is used **6 times**. [3]

Energy transferred = J

(ii) Use the equation:

$$\text{current} = \frac{\text{power}}{\text{voltage}}$$

to calculate the current when the hand drier is used. [2]

Current = A



Examiner
only

- (iii) The hand drier is connected to the mains with a standard three-core cable. Complete the table below with the value of the current in each wire in the connecting cable when the hand drier is operating normally. [2]

Wire	Current (A)
live	
neutral	
earth	

- (c) A fault develops and the live wire touches the metal case of the hand drier. Explain how a residual current circuit breaker (rccb) protects the user. [2]

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- (d) A salesperson for a replacement hand drier claims it will use fewer units of electricity each time it is used because its power is 1 300W. When it operates it blows hot air for double the time. Explain if you agree with the claim of the salesperson. [2]

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12



6. Water waves on the sea where the water is shallower than $\frac{1}{20}$ of their wavelength are known as shallow water waves. The speed of shallow water waves is described by the equation:

$$v = 3.13\sqrt{d}$$

where v is the wave speed (in m/s) and d is the depth of the water (in m).
This equation applies to sea waves whose wavelengths range between 10 m and 150 m.

In regions of the sea where the depth is small, for example near the shore, the speed noticeably changes but the frequency of the waves remains constant.

A shallow water wave is an example of a transverse wave.

- (a) Describe what is meant by a transverse wave. [2]

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- (b) (i) Use the equation above to **complete the table below**. [2]
Space for workings.

Depth of water, d (m)	$\sqrt{d}$	Wave speed, v (m/s)
0	0	0
0.5	0.71	2.21
1.0	1.00	3.13
1.5		3.83
2.5	1.58	4.95
3.0	1.73	5.42
3.5	1.87	
4.0	2.00	6.26



Examiner
only

- (ii) Chris suggests that if the depth of the water increases four times, the wave speed doubles. **Use data in the table opposite** to explain whether or not this statement is true. [2]

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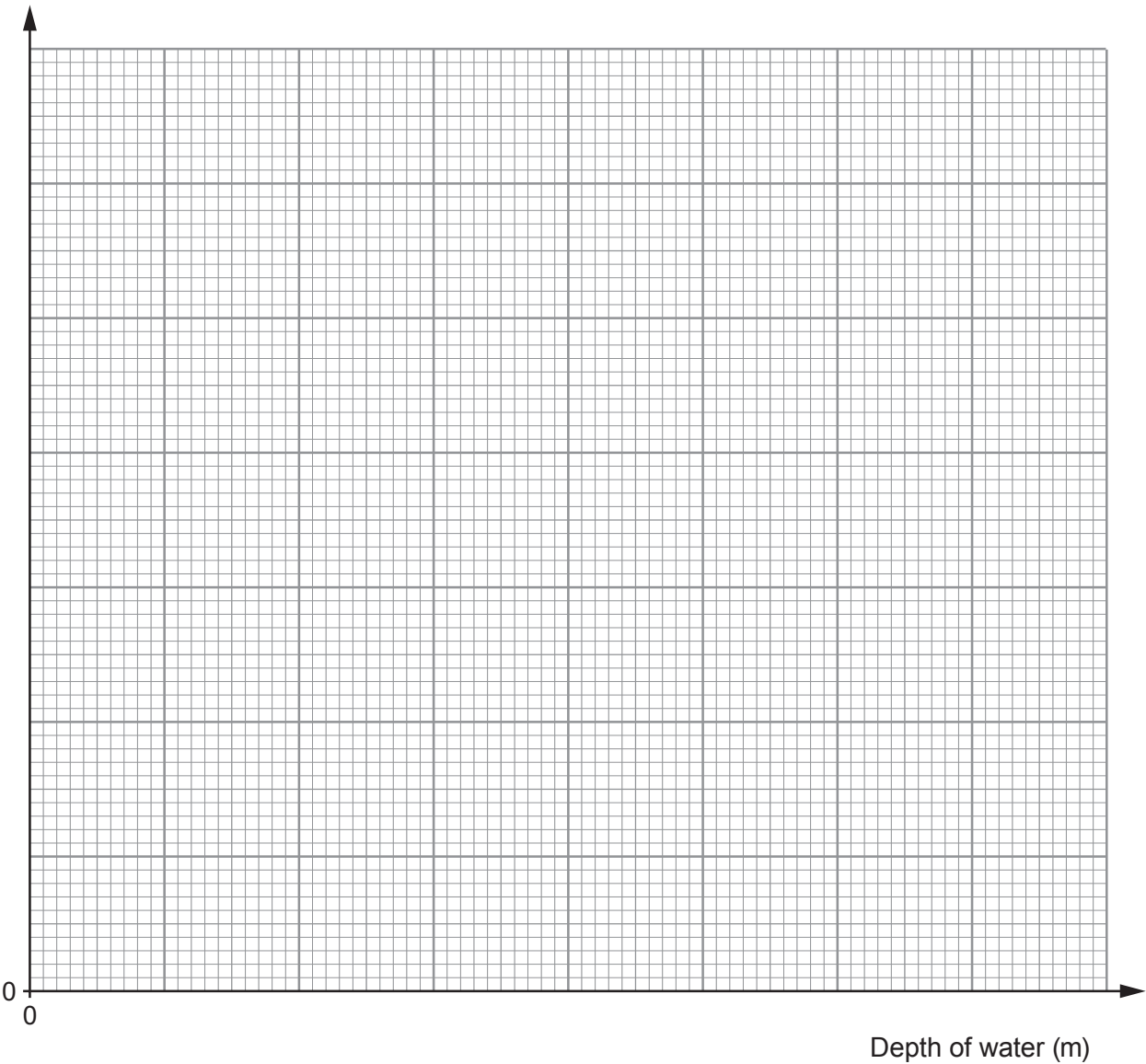
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- (iii) Plot the data on the grid below and draw a suitable line. [4]

Wave speed (m/s)



Examiner
only

(c) (i) Use the graph and the equation:

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

to calculate the wavelength of water waves that have a frequency of 0.2 Hz in water that is 2.0 m deep. [3]

Wavelength = m

(ii) Chris now suggests that as the depth increases, the wavelength decreases. Explain whether this statement is correct. [2]

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END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		0



GCSE
3430U30-1
SCIENCE (Double Award)



Unit 3 – PHYSICS 1
FOUNDATION TIER

FRIDAY, 14 JUNE 2019 – MORNING
1 hour 15 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	5	
3.	5	
4.	9	
5.	6	
6.	14	
7.	15	
Total	60	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

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Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	

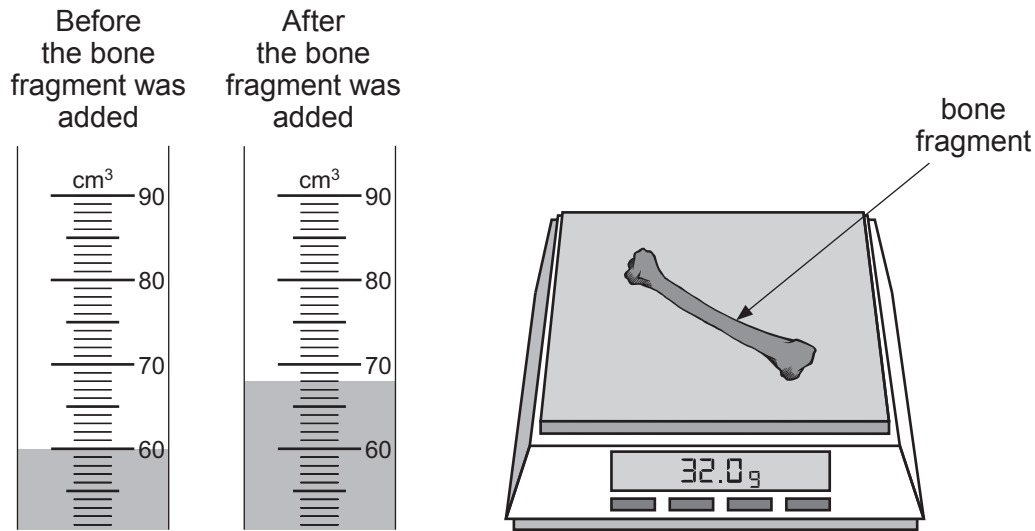
SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6



Answer all questions.

1. The diagrams below show the apparatus used to find the density of a fragment of bone. The diagram of the measuring cylinder shows the water level before and after the bone fragment was added.



- (a) (i) Use the information in the diagrams to **complete the table below**. [2]

Mass of bone	 g
Volume of bone	 cm ³

- (ii) Use an equation from page 2 to calculate the density of the bone fragment. [2]

Density = g/cm³

- (b) Explain, in terms of particles, why bone has a greater density than water. [2]

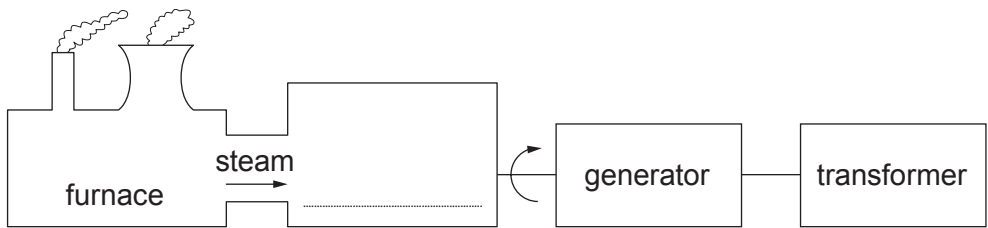
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2. The diagram below shows the main parts of a thermal power station that uses coal to produce electricity.



- (a) (i) **Complete the labelling on the diagram.** [1]
(ii) Draw **one** line from each part of the power station to the energy change it produces. [2]

	kinetic energy to electrical energy
furnace	
	chemical energy to electrical energy
generator	
	chemical energy to thermal (heat) energy

(b) Some energy values for this power station are listed below.

Input energy = 15 000 kJ
Heat energy = 10 500 kJ
Electrical energy = 4 500 kJ

Use this information and the equation below to calculate the % efficiency of the power station. [2]

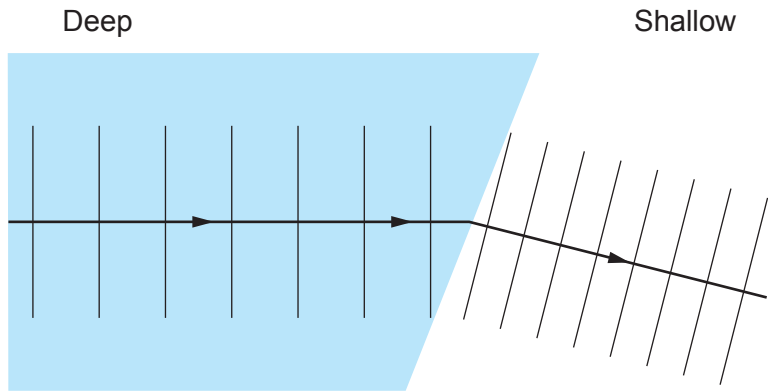
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$

% efficiency =



3. A ripple tank is used to demonstrate what happens to water waves as they travel from deep water into shallow water.

The results are shown in the diagram below.



- (a) The water waves change direction at the boundary between deep and shallow water.
State the name of this effect. [1]
- (b) Tick (✓) the boxes next to the **two** correct statements that describe the water waves. [2]
- | | |
|---|--------------------------|
| The frequency of the waves is higher in shallow water | ☐ |
| The amplitude of the waves is the same in deep and shallow water | ☐ |
| The wave speed is less in shallow water | ☐ |
| The wavelength decreases as the waves pass from deep to shallow water | ☐ |
- (c) In deep water the waves have a wavelength of 3 cm and a frequency of 5 Hz. Use this information and an equation from page 2 to calculate the wave speed. [2]

Wave speed = cm/s



4. The Government intends to phase out the use of petrol and diesel cars by 2040 and switch to electric vehicles. Two such electric vehicles are the Voltsa and the Ampra. The Voltsa has an electric motor only. The Ampra has both an electric motor and a petrol engine. The table gives information about these electric vehicles and a petrol only car.

	Voltsa	Ampra	Petrol only
Distance travelled on one tank of fuel (km)		450	600
Distance travelled on one charge (km)	150	60	
Cost of one full charge (p)	264	90	
Mean volume of petrol used (litres/100 km)	0	4	7
Cost of petrol (p/litre)		120	120
Mean CO ₂ produced over 100 km (g/100 km)	0	7 000	12 000

Use the information in the table to answer the following questions.

(a) (i) Eric travels 110 km each day. He is concerned about his carbon footprint. Why would he choose a Voltsa rather than an Ampra? [1]

.....

.....

(ii) Calculate the mass of CO₂ produced if the Ampra is driven 250 km. [2]

Mass of CO₂ = g

(b) (i) The Voltsa can be charged fully in six hours using a 4 kW charger. When 1 kWh of electricity is generated it produces 0.4 kg of carbon dioxide. Calculate how much CO₂ is produced to fully charge the Voltsa. [2]

Mass of CO₂ = kg

(ii) Ian says that the CO₂ data for the Voltsa is misleading. Explain why. [2]

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Examiner
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(c) Ian drives 600 km every week. He says he is not going to buy the more expensive **Volt**sa since it will not save him money to run compared to his **petrol only** car. Explain whether you agree with Ian. [2]

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5. Name the different regions found within the electromagnetic spectrum **and** describe how they are similar to **and** different from, one another. [6 QER]

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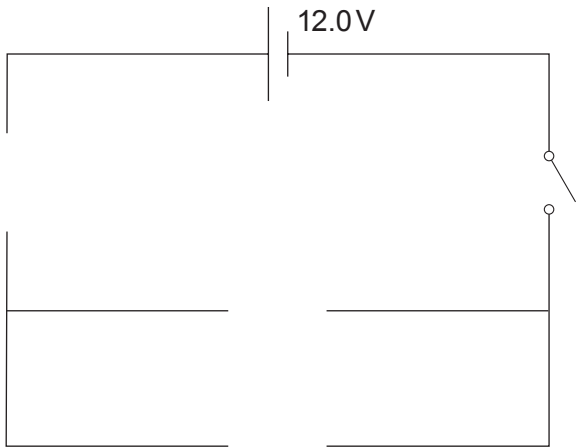
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6. A thermistor is a type of resistor whose resistance changes with temperature. Thermistors are used as temperature sensors.

To investigate a thermistor, students set up the following circuit to take measurements of current through the thermistor and the voltage across it.



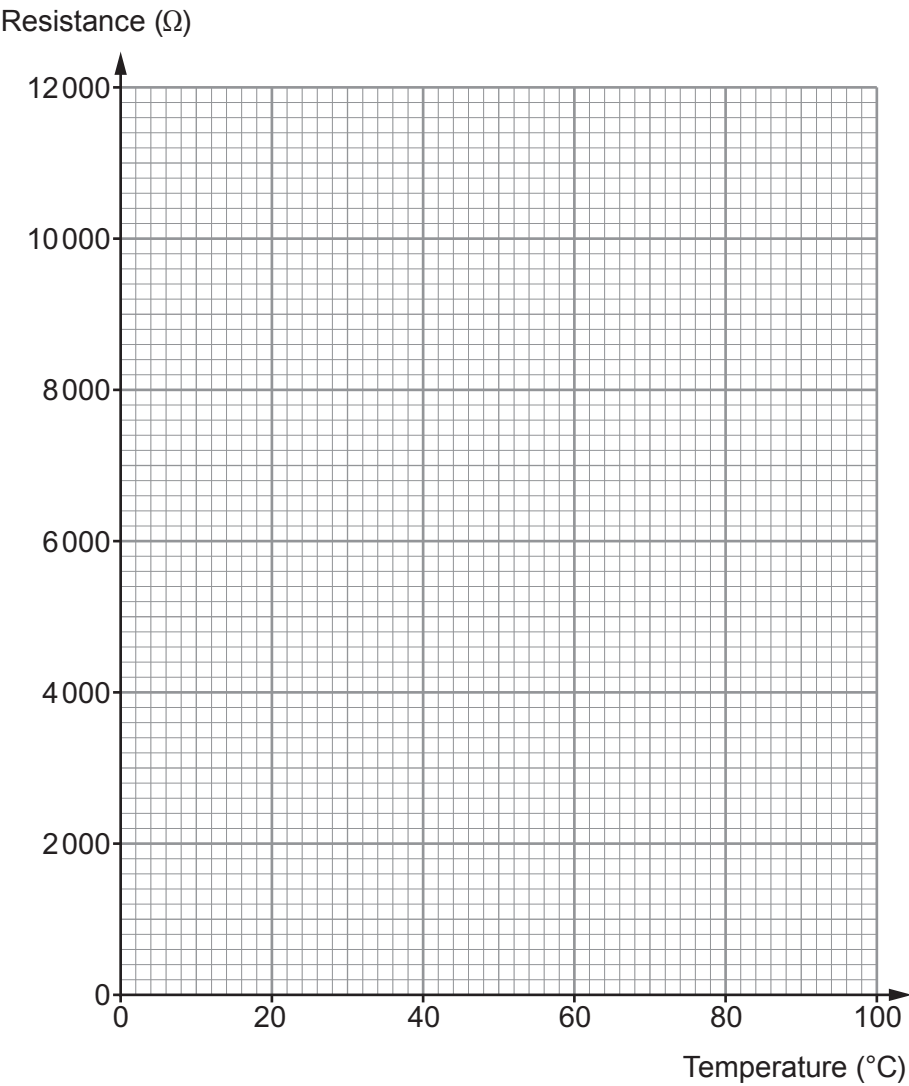
- (a) **Complete** the circuit diagram using the correct circuit symbols. [3]



(b) The students place the thermistor in water. They take current and voltage readings at different temperatures. These are used to calculate the resistance of the thermistor at each temperature. The results are shown in the table below.

Temperature (°C)	Current (mA)	Voltage (V)	Resistance (Ω)
20	1.0	12.0	12 000
40	2.2	12.0	5 400
60	4.8	12.0	2 500
80	8.6	12.0	1 400
100	20.0	12.0	600

(i) Plot the data on the grid below and draw a suitable line. [3]



Examiner
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(ii) Describe what happens to the resistance of the thermistor as the temperature increases. [2]

.....

.....

.....

(c) (i) Use your graph to find the resistance of the thermistor at 50 °C. [1]

Resistance at 50 °C = Ω

(ii) Use an equation from page 2 to calculate the current through the thermistor at 50 °C. [3]

Current = A

(d) A temperature sensor needs to vary in resistance by at least 3600 Ω as the temperature changes from 40 °C to 80 °C. Use the results from the experiment to explain whether the thermistor used by the students would be suitable. [2]

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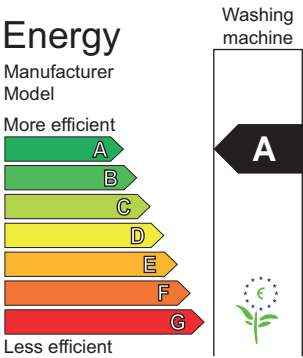
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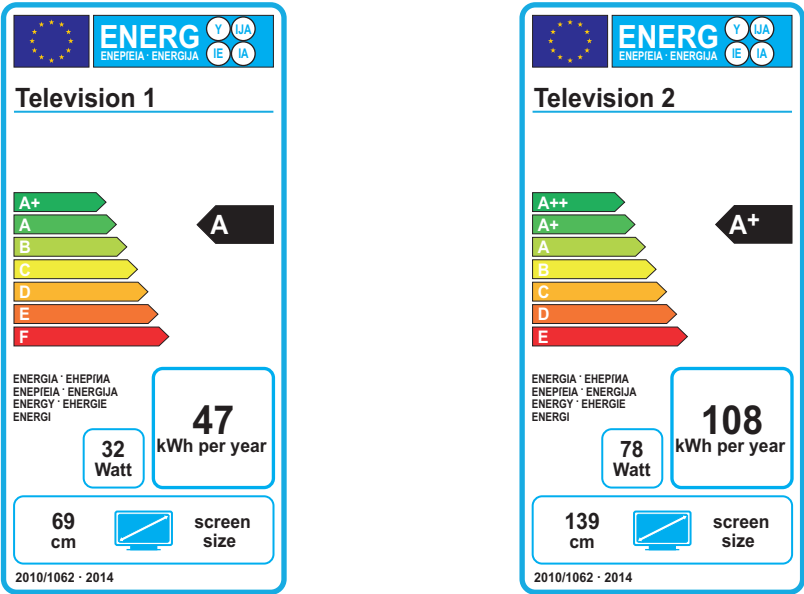


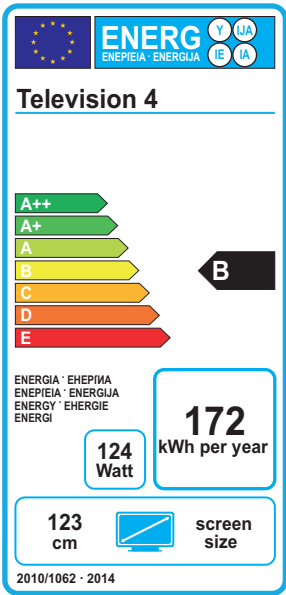
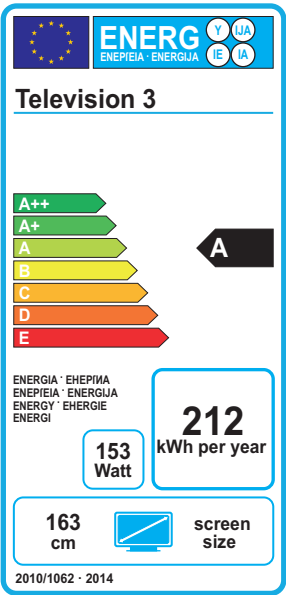
7. Energy rating labels are compulsory on most household appliances such as fridges, dishwashers, washing machines and televisions. These labels allow customers to compare appliances. In addition, the labels give other information about the appliances such as how noisy they are.
- Labels used to rate appliances are from A, the most efficient, to G, the least efficient.



As manufacturers have designed more efficient devices, new categories, e.g. A+, A++ and A+++ have been added to the labels.

The energy rating labels of four televisions are given below and opposite.





Information on the cost of each television is given below.

Television	Purchase cost (£)
1	280
2	1 000
3	1 500
4	800

(a) Use the information from the labels and the table to tick (✓) the **three** correct statements below. [3]

- Television 1 uses less energy per second than television 2 ☐
- The largest televisions always use the most energy ☐
- The purchase cost of television 2 is 1.5 times that of television 3 ☐
- More expensive televisions always use less energy ☐
- Television 3 uses 40 units more per year than television 4 ☐
- Televisions with the same energy rating, e.g A+, don't always have the same power ☐



- (b) It is claimed that power is proportional to screen size. Use the data for television **1** and television **2** to determine if this claim is true. [3]
Space for calculations.

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.....

- (c) (i) Use the equation:

$$\text{time (h)} = \frac{\text{units used (kWh)}}{\text{power (kW)}}$$

and data from the energy label to calculate how many hours the label suggests that television **2** is used for in 1 year. [2]

Hours used =

- (ii) Use an equation from page 2 to calculate the cost of using television **2** for 1 year if 1 unit (kWh) of electricity costs 16 p. Give your answer in pounds (£). [2]

Cost = £



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only

- (iii) The expected lifespan of a television is 10 years. Simon concludes that it will be more cost effective to buy and run television **2** for 10 years but Sarah disagrees and claims that television **4** will be cheaper. Use the data to determine who is right. [3]
Space for calculations.

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- (iv) Other than to save money, why should consumers be encouraged to choose appliances that use less energy? [2]

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END OF PAPER

15



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3420U10-1



MONDAY, 20 JUNE 2022 – MORNING

PHYSICS – Unit 1:
Electricity, Energy and Waves

FOUNDATION TIER

1 hour 45 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	12	
3.	6	
4.	13	
5.	12	
6.	7	
7.	7	
8.	13	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **5(a)**.



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Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	
pressure = $\frac{\text{force}}{\text{area}}$	$p = \frac{F}{A}$
change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature	$\Delta Q = mc\Delta\theta$
thermal energy for a change of state = mass $\times$ specific latent heat	$Q = mL$
V_1 = voltage across the primary coil V_2 = voltage across the secondary coil N_1 = number of turns on the primary coil N_2 = number of turns on the secondary coil	$\frac{V_1}{V_2} = \frac{N_1}{N_2}$

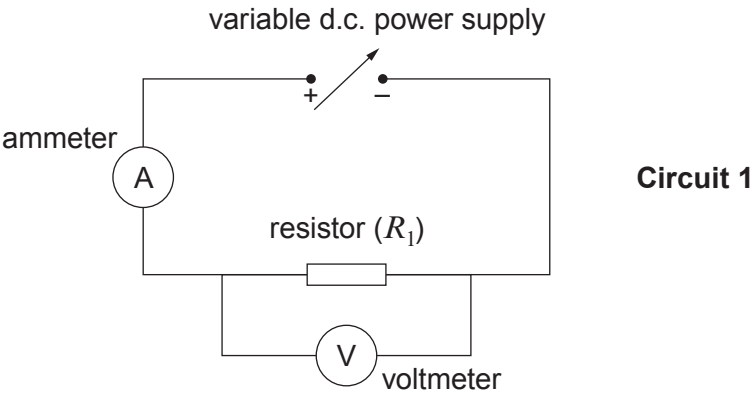
SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6

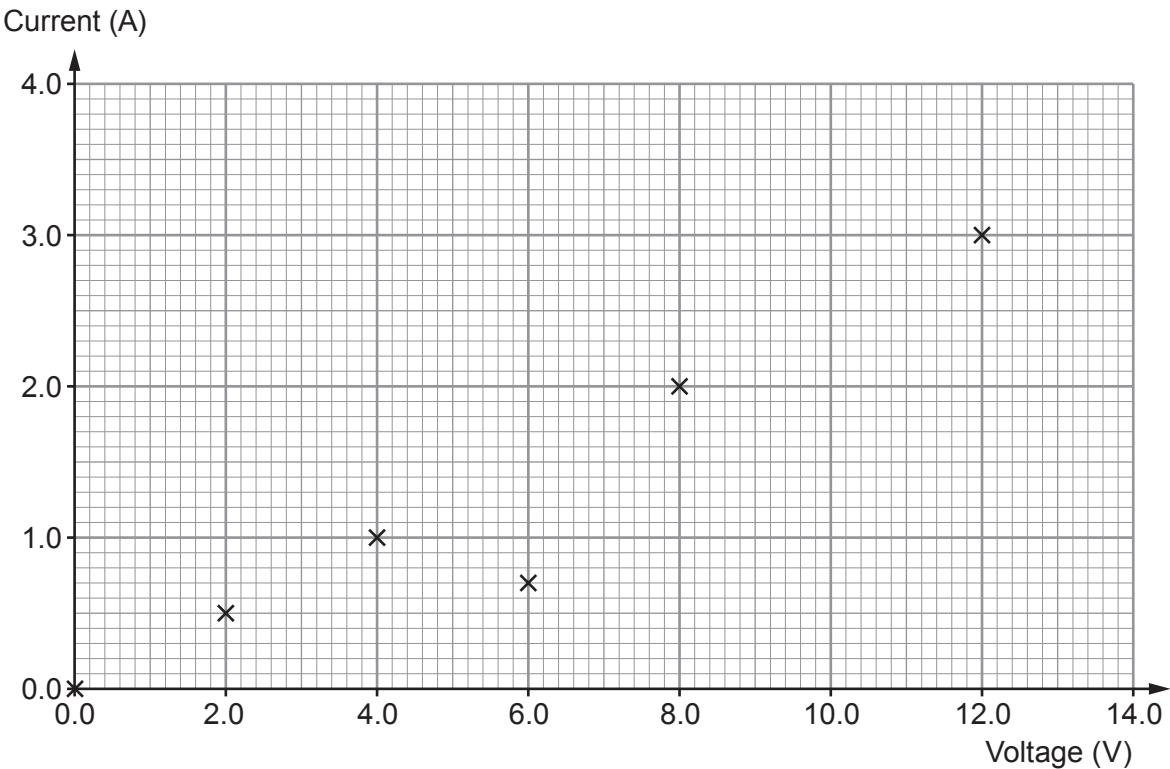


Answer **all** questions.

1. Paul investigated how the current changes with voltage for a resistor (R_1). The circuit he used is shown (**Circuit 1**).



- (a) Name the device he used to measure the current. [1]
- (b) Paul's results are shown on the graph.



- (i) On the graph, **circle** the anomalous point. [1]
- (ii) **Complete the graph** by drawing a suitable straight line. [1]



(iii) I. Use the graph to state the current at 12.0 V. [1]
Current = A

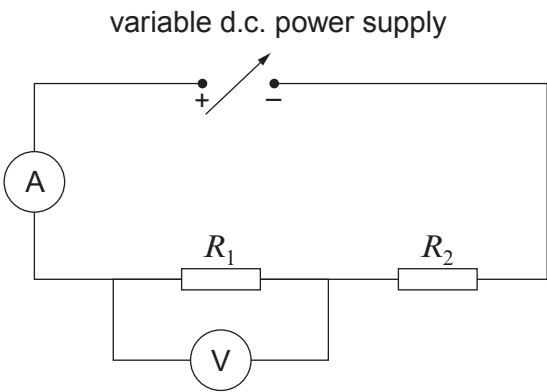
II. Use the equation:
$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the resistor (R_1) at 12.0 V. [2]

Resistance of R_1 = Ω

(iv) State what the graph shows about the resistance of the resistor (R_1) as the voltage changes. [1]

(c) The original circuit is then adapted by adding a second, **identical** resistor (R_2) in series with R_1 .
This circuit is shown below (**Circuit 2**).



Complete the following sentences by underlining the correct word or phrase in each bracket. [3]

The total resistance in circuit 2 is (**double / the same as / half**) the total resistance of circuit 1.

When the d.c. power supply is set at the same voltage, the voltmeter reading in circuit 2 is (**double / the same as / half**) the voltmeter reading in circuit 1.

The current in circuit 2 is (**double / the same as / half**) the current in circuit 1.

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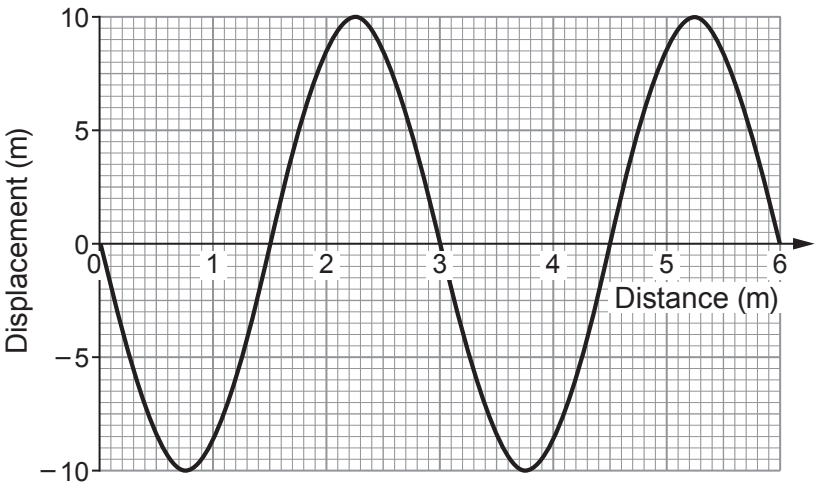
2. The diagram shows regions of the electromagnetic (em) spectrum.
The regions are in order.

radio waves	microwaves	infra-red	visible light	ultraviolet	X-rays	gamma rays
-------------	------------	-----------	---------------	-------------	--------	------------

(a) **Tick (✓) the *three* correct statements.** [3]

- The frequency of em waves decreases from radio waves to gamma rays.☐
- All em waves travel at the same speed in a vacuum.☐
- Gamma rays can be emitted from radioactive materials.☐
- The wavelength of em waves decreases from radio waves to gamma rays.☐
- Radio waves, microwaves, infra-red and visible light are all ionising radiations.☐
- All em waves are longitudinal.☐

(b) The graph below shows a wave diagram.



Use the following values to answer the questions below. You may use each value once, more than once or not at all.

2	3	5	6	10
---	---	---	---	----

- (i) I. State the amplitude of the wave. m [1]
- II. State the number of complete waves shown. waves [1]
- III. State the wavelength of the wave. m [1]



(ii) I. Complete the following sentences by underlining the correct phrase in each bracket. [2]

The wave has a frequency of (7.5Hz / 7.5cm/s / 7.5cm).

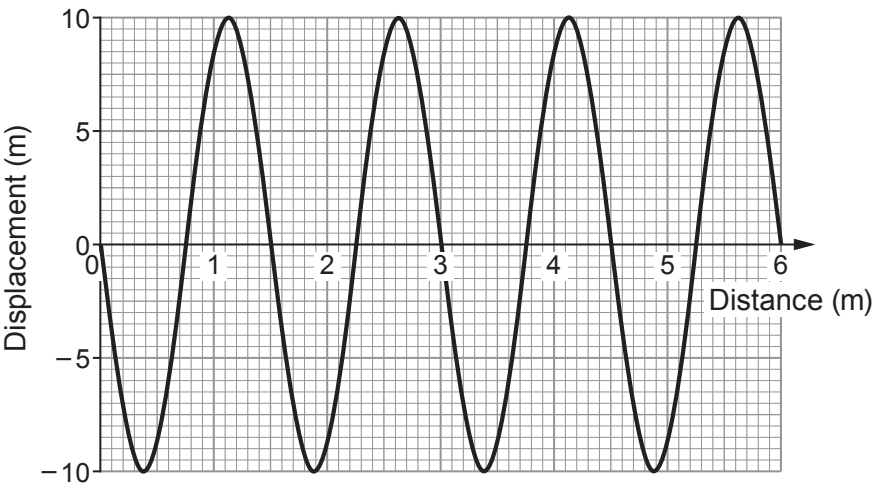
This means 7.5 waves pass a point each (second / minute / hour).

II. Use information from parts (b)(i) and (ii) to calculate the wave speed.
Use the equation:

wave speed = wavelength $\times$ frequency [2]

Wave speed = m/s

(c) The speed of the wave remains constant but its **frequency is doubled**.
A student draws a diagram of the expected wave. It is shown below.



Explain whether you agree with the student's diagram of the wave. [2]

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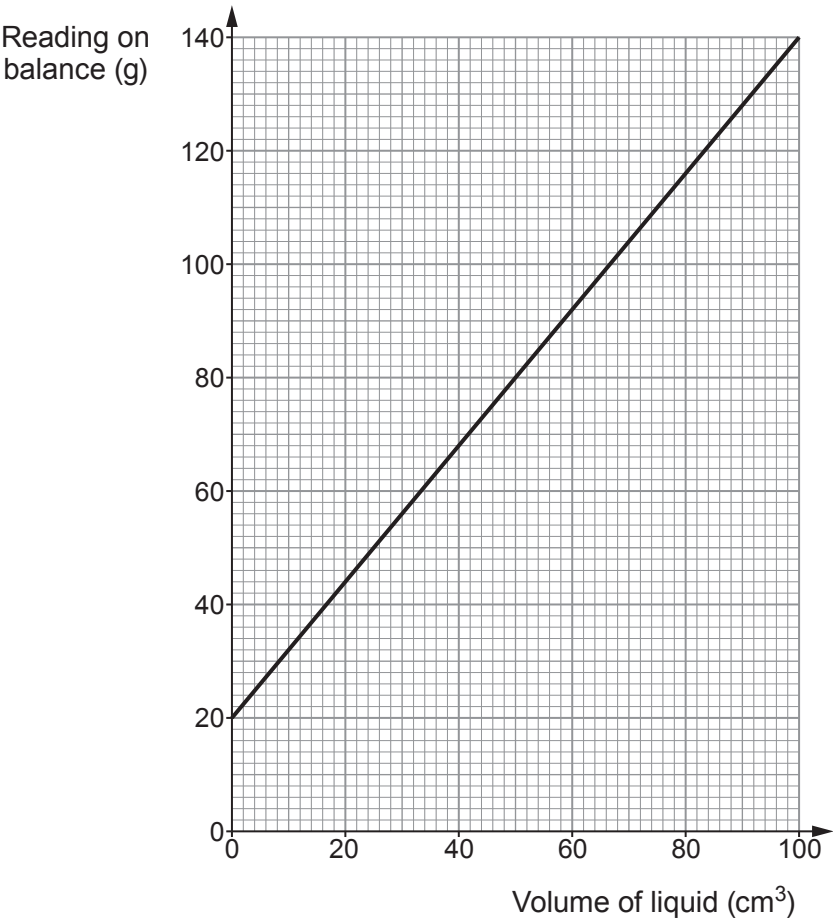
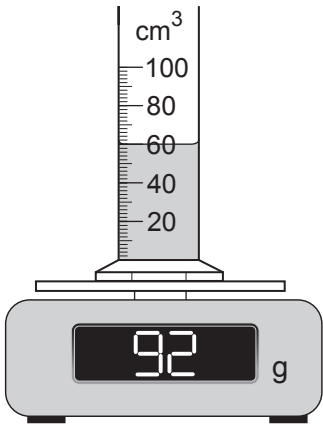
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3. Kylie carries out an experiment, in class, to measure the density of a liquid. She places an empty measuring cylinder on a balance and pours the liquid into it. She records the reading from the balance for different volumes of liquid. Her results are shown in the table and on the graph.

Volume of liquid (cm ³)	Reading on balance (g)
20	44
40	68
60	92
80	116
100	140



Use the information above to answer the following questions.

- (a) State the mass of the empty measuring cylinder. g

[1]



Examiner
only

(b) Calculate the mass of liquid in the measuring cylinder when its volume is 100 cm³. [2]

Mass of liquid = g

(c) Use an equation from page 2 to calculate the density of the liquid that Kylie used and state the unit of your answer. [3]

Density =

Unit =

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6



4. A Welsh farmer decides to purchase a wind generator to provide some electrical energy for his farm.
A selection of wind generators is available.

	Wind generator 1	Wind generator 2	Wind generator 3
Cost to buy and install (£)	1 800	3 000	2 925
Expected lifetime (years)	15	15	15
Estimated number of units produced per year (kWh)	2 000	4 000	3 000
Cost per unit (£)	0.15	0.15	0.15
Estimated saving per year (£)		600	450
Estimated payback time (years)		5.0	6.5

- (a) (i) Use the equation:

$$\text{estimated saving per year} = \text{units produced per year} \times \text{cost per unit}$$

and data from the table to calculate the estimated saving per year if **wind generator 1** was installed.

[1]

Estimated saving per year = £

- (ii) Use the equation:

$$\text{estimated payback time} = \frac{\text{cost to buy and install}}{\text{estimated saving per year}}$$

and data from the table to calculate the estimated payback time for **wind generator 1**.

[2]

Estimated payback time = years



- (b) (i) A salesperson suggests to the farmer that **wind generator 3** is better than **wind generator 2** because it is cheaper.
Explain, using information from the table, if the salesperson’s suggestion is correct over a 5-year timescale. [2]

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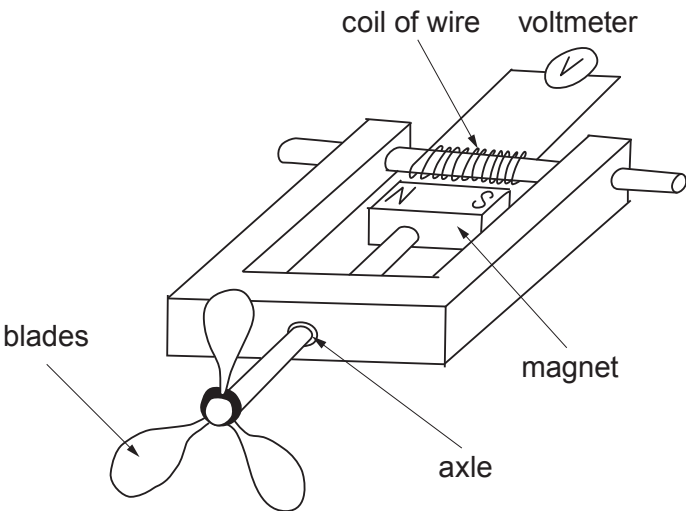
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- (ii) Give **two** reasons why the payback time can only be an estimate. [2]

1.
2.

- (c) A wind generator only operates in certain wind speeds.
The diagram below shows a student’s design for measuring wind speed.



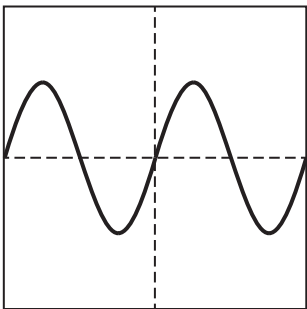
- (i) When the wind blows a reading on the voltmeter is obtained.
The following sentences describe the different stages.
- A** The magnetic field lines cut through the coil of wire.
 - B** The blades rotate.
 - C** The magnet rotates.
 - D** This induces a voltage in the coil of wire.

Put the letters **A**, **B** and **C** in the boxes to correctly describe the process of inducing a voltage in the coil of wire. The final box has already been completed, [2]

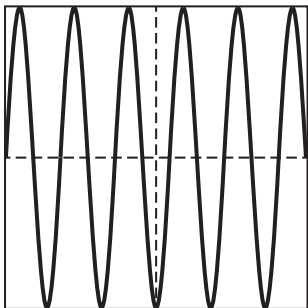
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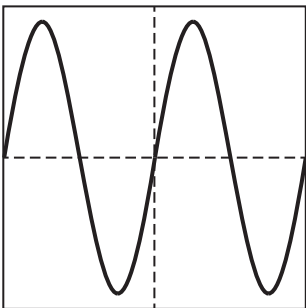
(ii) When the wind speed is **2 m/s** the alternating voltage produced is shown below.



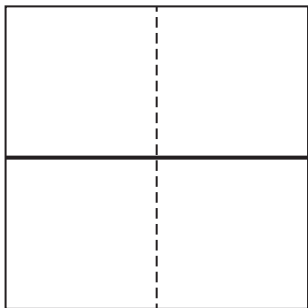
The following traces **X**, **Y** and **Z** are produced during different wind conditions.



Trace **X**



Trace **Y**



Trace **Z**

I. State which trace (**X**, **Y** or **Z**) is produced when the wind speed is **faster than 2 m/s**.

Trace [1]

II. State which trace (**X**, **Y** or **Z**) is produced when **the wind stops**.

Trace [1]

(d) When the student's design is built it is found that the voltmeter is not sensitive enough to measure a wind speed less than 0.5 m/s.

(i) Suggest **one** change that can be made to the **magnet** so that wind speeds less than 0.5 m/s can be measured by the voltmeter. [1]

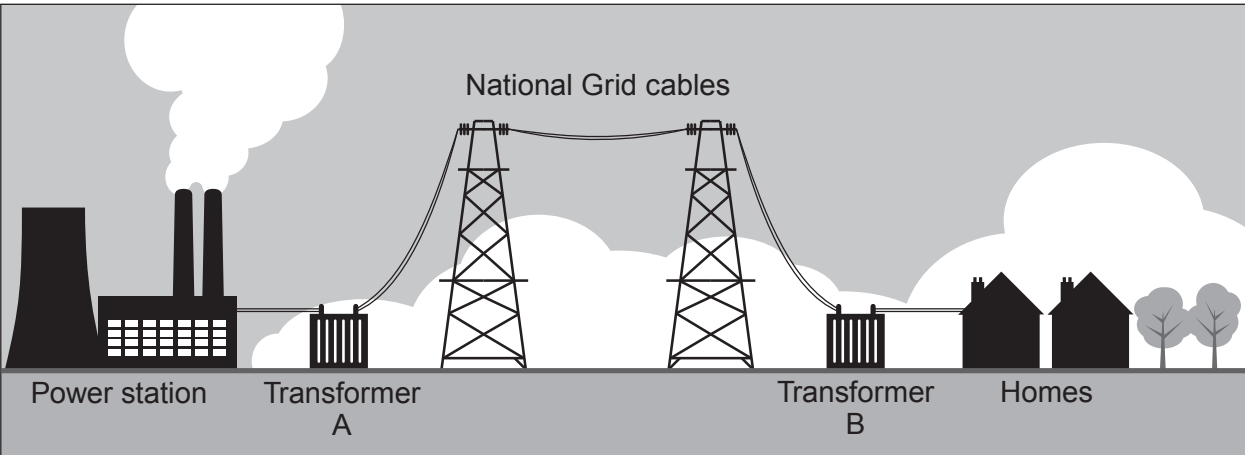
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(ii) Suggest **one** change that can be made to the **coil of wire** so that wind speeds less than 0.5 m/s can be measured by the voltmeter. [1]

.....



5. The National Grid includes a large network of cables that connects power stations to homes.



(a) Describe the purpose of transformers A and B in the National Grid.

Include in your answer:

- how electrical energy is transmitted **efficiently**
- how electrical energy is transferred **safely** to homes.

[6 QER]

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Examiner
only

- (b) Some houses have solar panels on their roof.
They produce electricity.
Explain why these houses are still connected to the National Grid. [2]

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- (c) A housing development of 20 new houses needs to be connected to the National Grid.
Each house requires a **maximum current of 60 A**.

- (i) Calculate the total current needed if all 20 houses are using the maximum current. [1]

Total current = A

- (ii) The mains voltage in the houses is 230 V a.c.

I. Use the equation:

power = voltage × current

to calculate the maximum total power supplied to the 20 houses. [2]

Maximum power = W

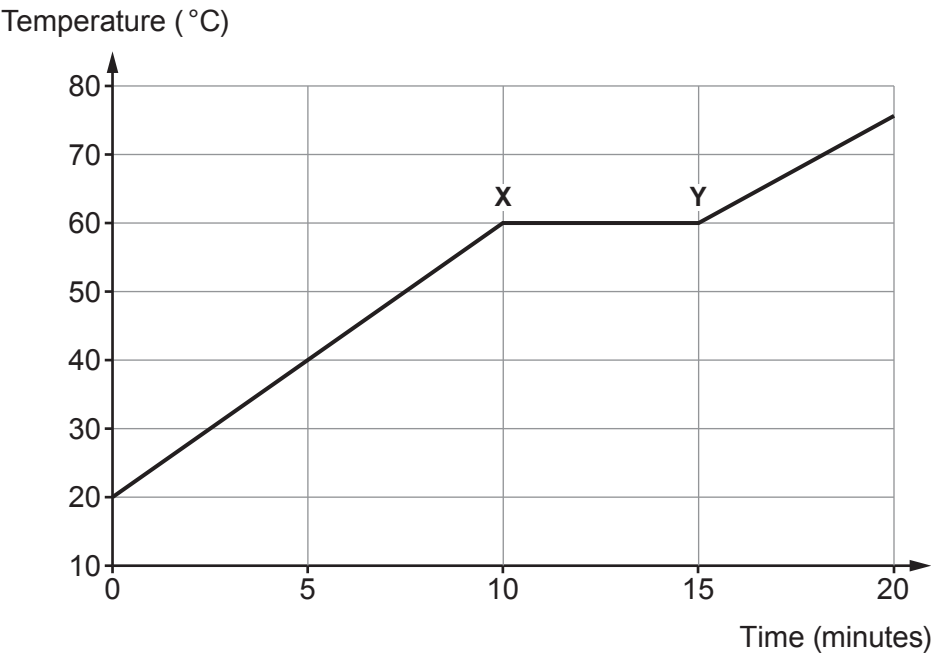
- II. Convert your answer for the maximum power into kW. [1]

Maximum power = kW

12



6. The graph below shows the results obtained when **36 g of solid wax** was heated using a heater. The temperature of the wax was recorded every minute.



(a) Explain why the energy supplied from the heater between **points X and Y** doesn't produce a rise in temperature of the wax. [2]

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(b) (i) Use the equation:

thermal energy supplied = mass $\times$ specific latent heat

to calculate the amount of thermal energy supplied to the 36 g of wax between **points X and Y**. [2]

(Specific latent heat of fusion (L_f) for wax = 220 J/g)

Thermal energy supplied = J



Examiner
only

(ii) All the energy is transferred from the heater to the wax during the 5 minute interval between **X and Y**.

I. Calculate the time, in seconds, of the interval between **X and Y**. [1]

Time = s

II. Use the equation:

$$\text{power} = \frac{\text{energy transferred}}{\text{time}}$$

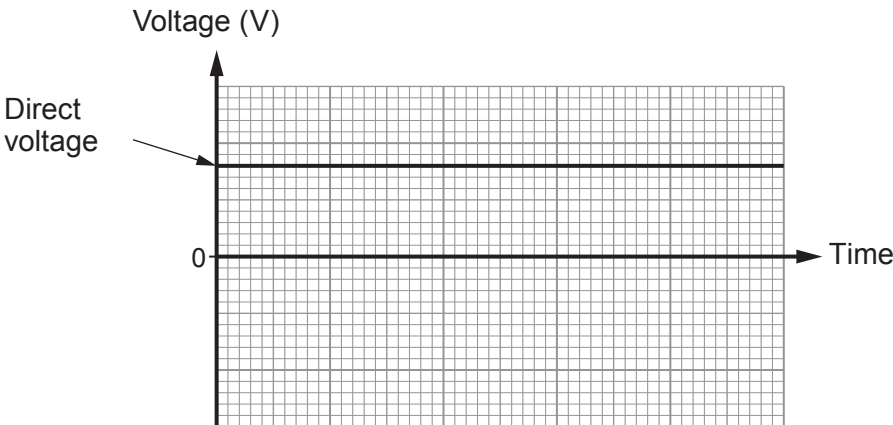
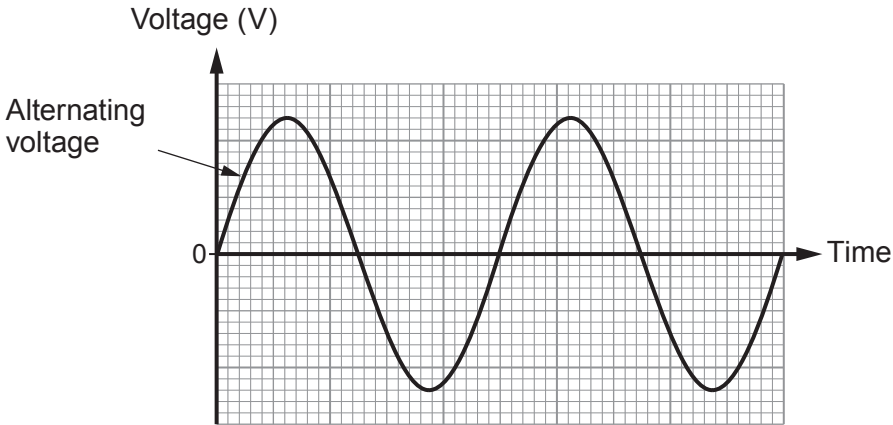
to calculate the power of the heater. [2]

Power of heater = W

7



7. (a) The mains electricity supply to homes is at a voltage of 230 V a.c.
In contrast to this, batteries supply d.c.
These are each shown in the diagrams below.



An alternating voltage changes in size, but a direct voltage keeps the same size.
State **one** other difference between the two.

[1]

.....



Examiner
only

- (b) Household circuits may be protected by a series of devices. These include:
- a miniature circuit breaker (mcb)
 - a fuse in a plug
 - a residual current circuit breaker (rccb).

The list below gives three situations in which one or another of these devices stop the power from being supplied.

1. An electric lawn mower cuts through the cable.
 2. Too much current flows in a ring main because too many appliances are being used at the same time.
 3. The live wire in a kettle touches its neutral wire.
- (i) Select and explain one situation above that puts the **user** in danger of an electric shock **and** which one of the devices should be used to keep the user safe. [2]

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- (ii) State **two** advantages of an mcb over a fuse in a household circuit. [2]
1.
 2.

- (c) Describe the function of the live and neutral wires in household circuits. [2]

Live wire:

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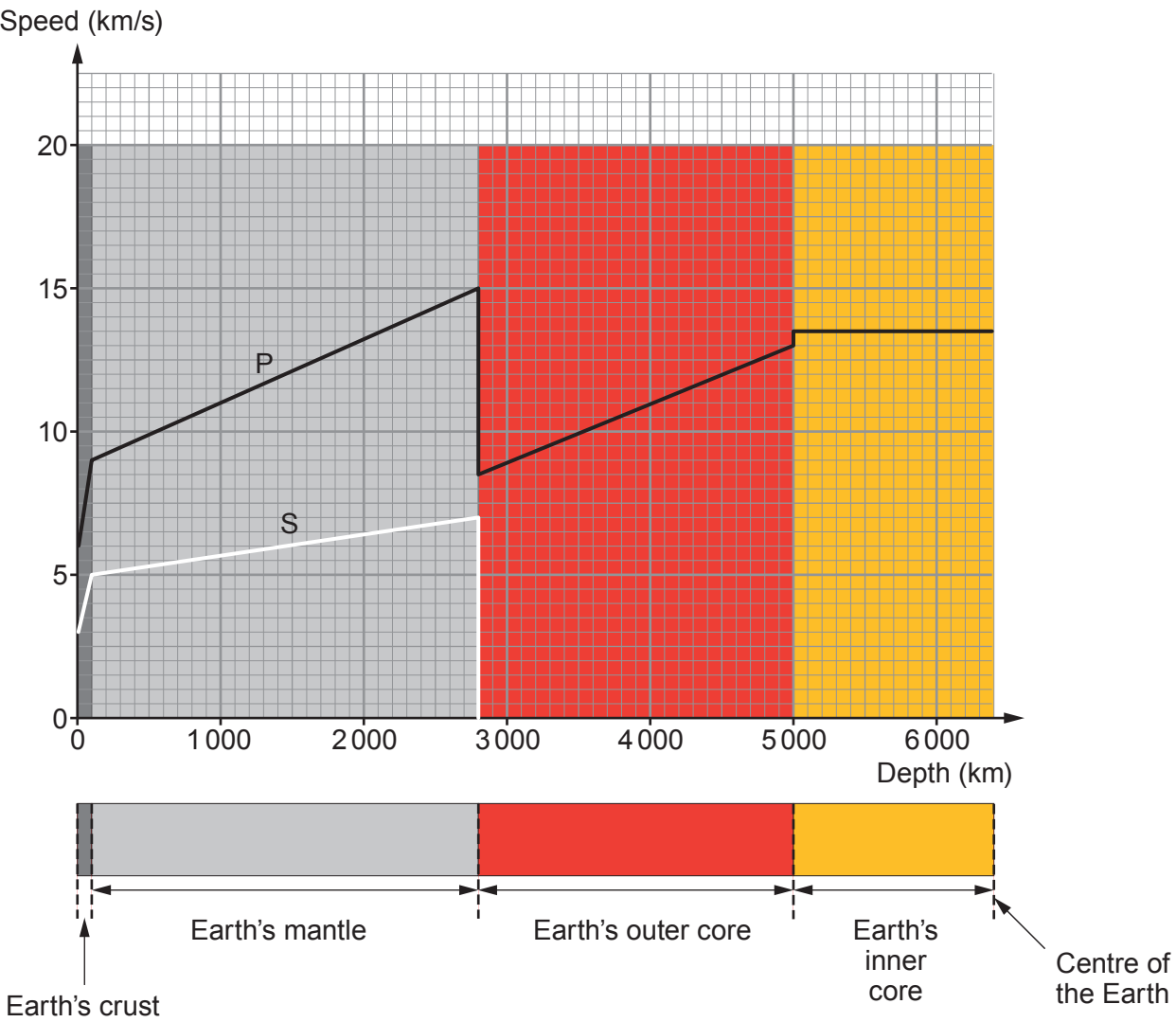
Neutral wire:

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8. P and S waves are produced during an earthquake. They travel at different speeds. The graph shows how the speed of P and S waves change with depth below the Earth's surface.



(a) P waves are longitudinal waves. Explain what is meant by a longitudinal wave. [2]

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.....

(b) Use the graph to calculate the difference in speed between the P wave and the S wave at a depth of **100 km**. [2]

Speed difference = km/s



- (c) (i) Bob claims that the maximum speed of P waves is greater in solids than in liquids. Explain, using data from the graph, whether Bob's claim is correct. [2]

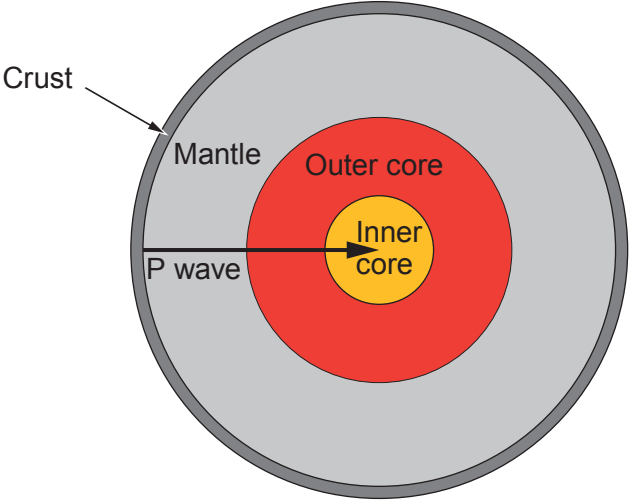
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- (ii) A P wave (shown in Diagram 1) travels a distance of 6 300 km in a straight line to the centre of the Earth in a time of 550 s.

Diagram 1



Bob states that the mean speed of the P wave over this distance is equal to its actual speed only at the depth of 3 500 km. Use an equation from page 2 to show whether Bob's statement is correct. [3]
Space for calculations.

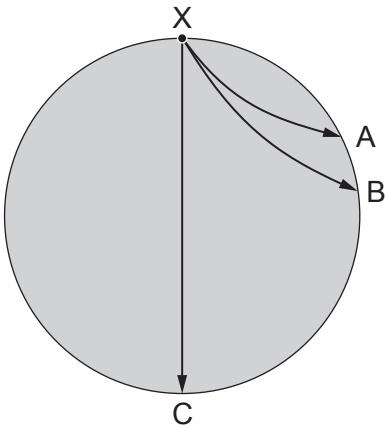
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(d) Diagram 2 shows waves passing through the Earth following an earthquake at point X.

Diagram 2



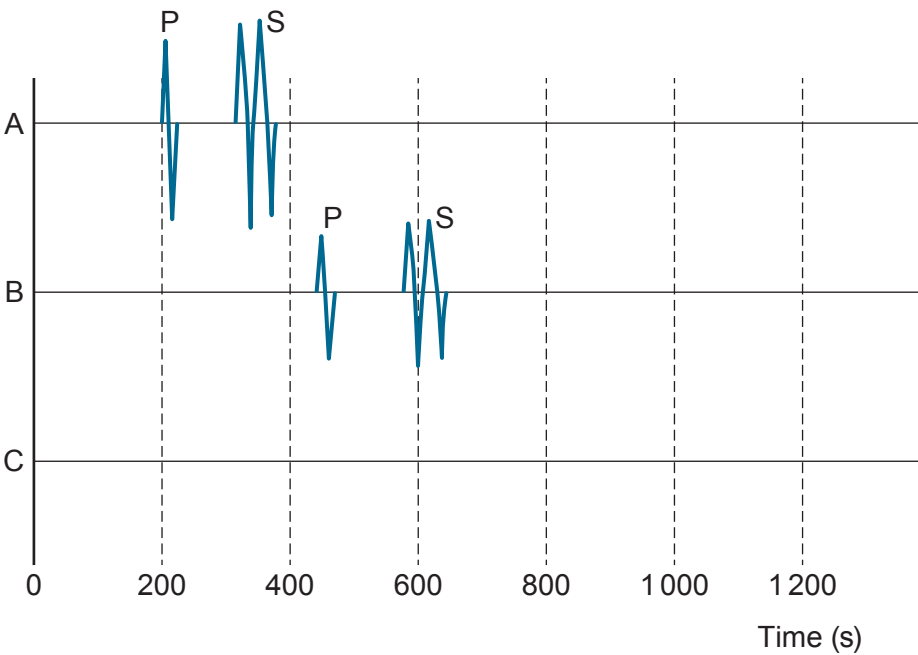
Monitoring stations at A, B and C detect the earthquake by signals that they receive.

- (i) Use the time given in part (c)(ii) to state the time taken for the P wave to travel from **X to C** (ignore the Earth's crust in this). [1]

Time = s

- (ii) The signals received at monitoring stations A and B are shown in Diagram 3. **Add to Diagram 3** the signal that you would expect to be received at station C. [3]

Diagram 3



END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3430U30-1



MONDAY, 19 JUNE 2023 – AFTERNOON

SCIENCE (Double Award)
Unit 3 – PHYSICS 1

FOUNDATION TIER

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	13	
3.	6	
4.	10	
5.	10	
6.	9	
7.	6	
Total	60	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **5(a)**.



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Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	

SI multipliers

Prefix	Symbol	Conversion factor	Multiplier
milli	m	divide by 1000	1×10^{-3}
centi	c	divide by 100	1×10^{-2}
kilo	k	multiply by 1000	1×10^3
mega	M	multiply by 1 000 000	1×10^6



Answer **all** questions.

1. The diagram shows the 7 regions of the electromagnetic (em) spectrum in order.



gamma rays	A	B	visible light	infra-red	microwaves	radio waves
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(a) Use **only** the words given in the box below to answer the following questions.

cosmic rays	X-rays	ultrasound	water	ultraviolet
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(i) Name the region **labelled A**. [1]

(ii) Name the region **labelled B**. [1]

(b) (i) Complete the following sentence by underlining the correct word in the bracket. [1]

The arrow shown on the electromagnetic spectrum above represents increasing (**amplitude** / **wavelength** / **frequency**).

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(ii) Waves in the electromagnetic spectrum all travel at the same speed in a vacuum.

Tick (✓) the **three** correct statements that describe other properties of em waves. [3]

☐

All transverse.

☐

All ionising.

☐

All transfer energy.

☐

All longitudinal.

☐

All travel through space.

☐

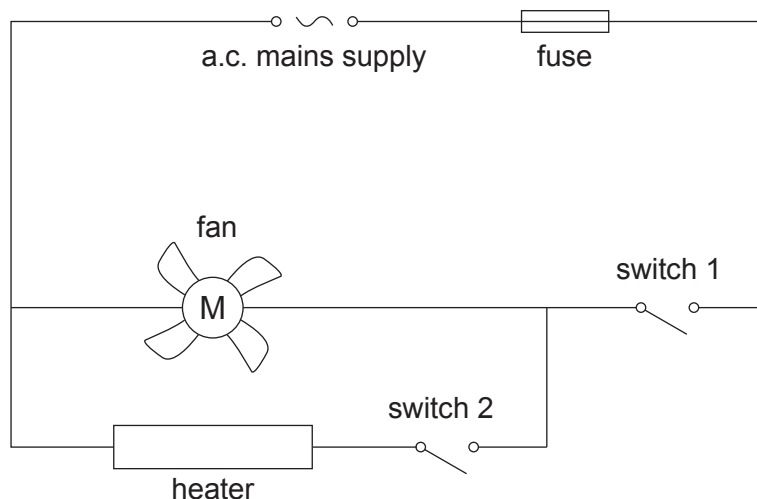
All given out by radioactive materials.

6



2. The circuit diagram below shows a design for a hairdryer that can blow either cold air or hot air.

The switches in the diagram are in the open position.



- (a) Complete the following sentences by underlining the correct phrase in the brackets. [2]

When switch 1 is closed and switch 2 is open the hairdryer blows cold air because
(only the fan is on / only the heater is on / the heater and fan are both on).

The hairdryer blows hot air when the heater and fan are both on. This happens when
(both switches are open / only switch 2 is closed / both switches are closed).

- (b) The hairdryer is used to blow hot air.

The current supplied is **6 A** and the mains voltage is 230 V.

- (i) Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power of the hairdryer.

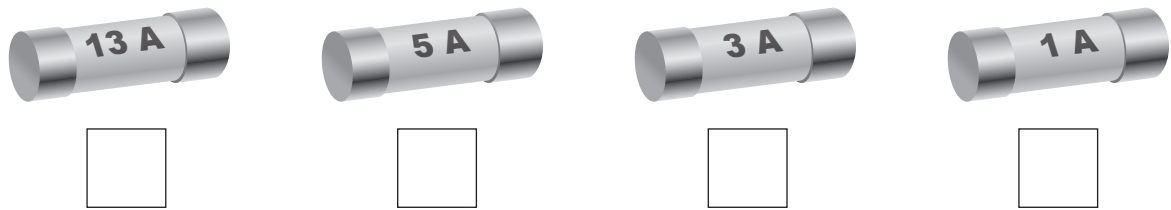
[2]

power = W



(ii) The following fuses are available.

Tick (✓) **the box** to show the correct fuse that should be used for the hairdryer. [1]



(iii) Complete the following sentence by underlining the correct phrase in the brackets. [2]

Fuses switch off circuits if the
(**current is too high** / voltage is too high / power is too low)
to prevent (**electric shocks** / wires overheating / power cuts).

(c) A **different hairdryer** is now used. The power of this hairdryer is **1.5 kW**.

(i) Use the equation:

$$\text{units used (kWh)} = \text{power (kW)} \times \text{time (h)}$$

to calculate the units used (kWh) by this hairdryer in **2 hours**. [2]

units used = kWh



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- (ii) The cost per unit of electrical energy is 30 p.
Use your answer to part (c)(i) and the equation:

cost = units used × cost per unit

to calculate the cost of using this hairdryer for 2 hours. [2]

cost = p

- (iii) State the cost of using this hairdryer for **double** the time. [1]

cost = p

- (iv) Tick (✓) **the box** that correctly shows the power of the 1.5 kW hairdryer in watts (W). [1]

- | | |
|--------------------------|-------------|
| ☐ | 1 500 000 W |
| ☐ | 1 500 W |
| ☐ | 1.5 W |
| ☐ | 0.015 W |

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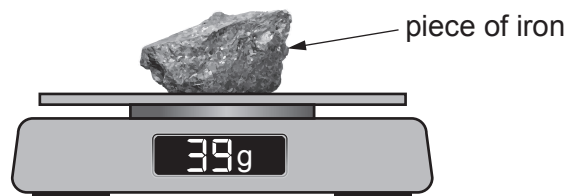
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3. Peter carries out an experiment to find the density of iron.

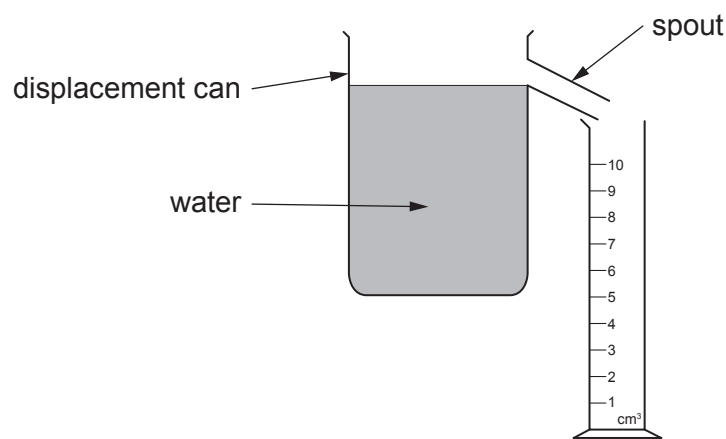
Stage 1

- He uses a balance to measure the mass of a piece of iron.



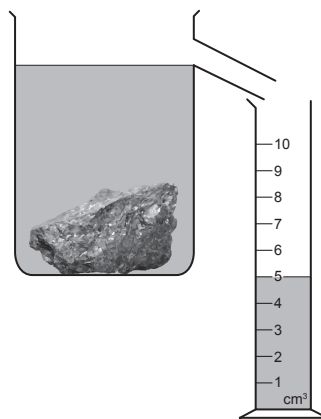
Stage 2

- He fills a displacement can with water.
- He waits until water stops flowing from the spout.
- He then places an empty measuring cylinder under the spout.



Stage 3

- He carefully lowers the iron into the water.
- He collects the water displaced in the measuring cylinder.



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Use information from the diagrams to answer the following questions.

(i) State the mass of the iron. mass = g [1]

(ii) State the volume of the iron. volume = cm³ [1]

(iii) I. Use the equation:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

to calculate the density of iron. [2]

density =

II. Underline the correct unit that should be used for the density above. [1]

g/cm³ **g/cm²** **cm²/g** **cm³/g**

(iv) State **one** change to the apparatus that would improve the accuracy of the measurement of the **mass** of the iron. [1]

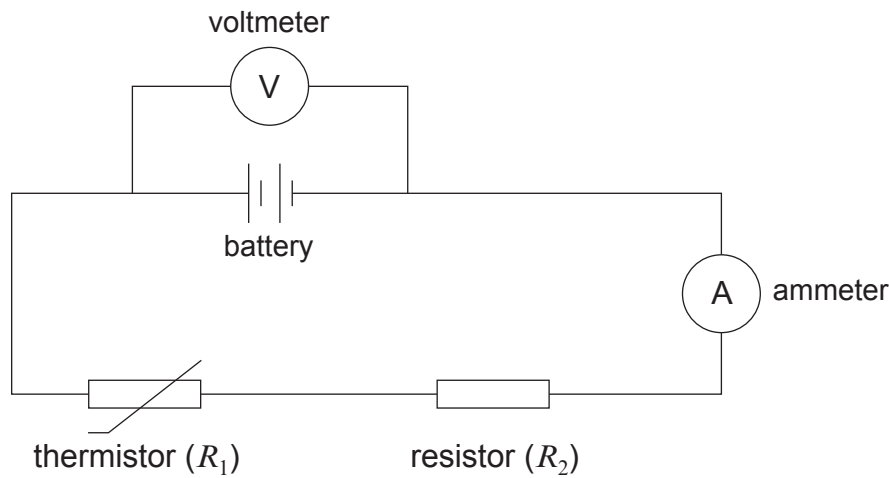
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4. Students set up the temperature sensing circuit shown below.



(a) At a temperature of 20 °C the resistance of the thermistor (R_1) is 300 Ω .

The resistor (R_2) has a fixed resistance of 50 Ω .

(i) Use the equation:

$$R = R_1 + R_2$$

to calculate the total resistance (R) of the circuit at 20 °C. [1]

total resistance = Ω



(ii) A photograph of the voltmeter used in the circuit is shown below.



State the voltage shown on the voltmeter. [1]

voltage = V

(iii) Use an equation from page 2 and your answers from parts (i) and (ii) to calculate the current in the circuit. [2]

current = A

(iv) State the ammeter reading. [1]

ammeter reading = A



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(b) Complete the following sentences by underlining the correct phrase or word. [3]

As the (**temperature** / **power** / **voltage**) of the thermistor is increased to 50 °C its resistance decreases. The total resistance in the circuit (**increases** / **stays the same** / **decreases**) and the circuit current (**increases** / **stays the same** / **decreases**).

(c) (i) State how the original circuit could be changed to become a light sensing circuit. [1]

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(ii) Give a reason for your answer. [1]

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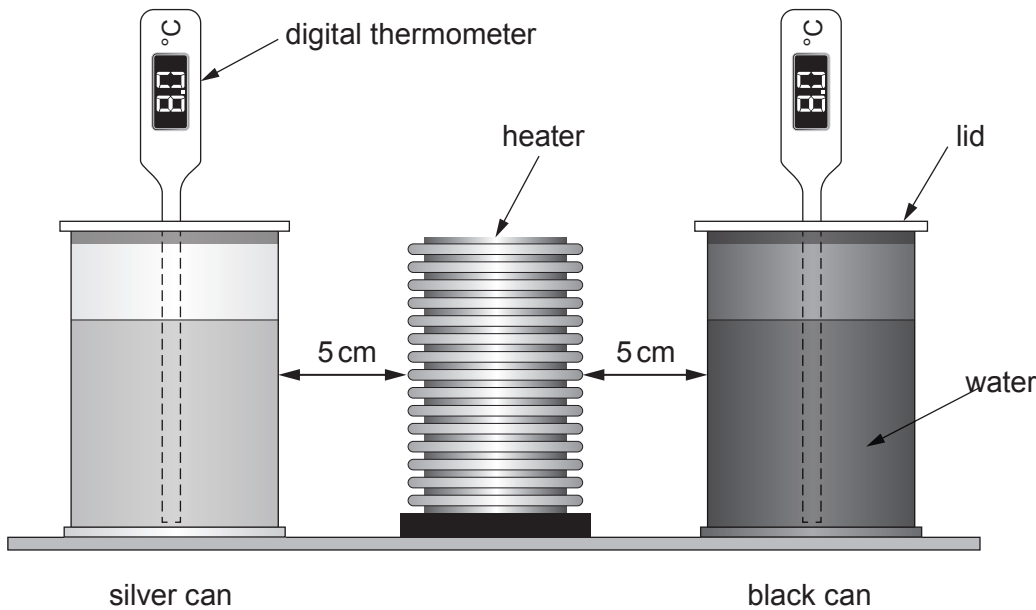


5. A pupil sets up an experiment to investigate the absorption of heat radiation.

The diagram shows the apparatus set up **before** the heater was turned on for 8 minutes.

Two identical aluminium cans were both filled with 200 cm³ of water and a digital thermometer was inserted through a small hole in the lid.

One can was painted black and the other painted silver.



(a) In the experiment several variables were controlled in order to make a valid conclusion.

Use the information above to describe how this was achieved.

In your answer include

- **three** variables that were controlled
- an explanation of why each was controlled.

[6 QER]

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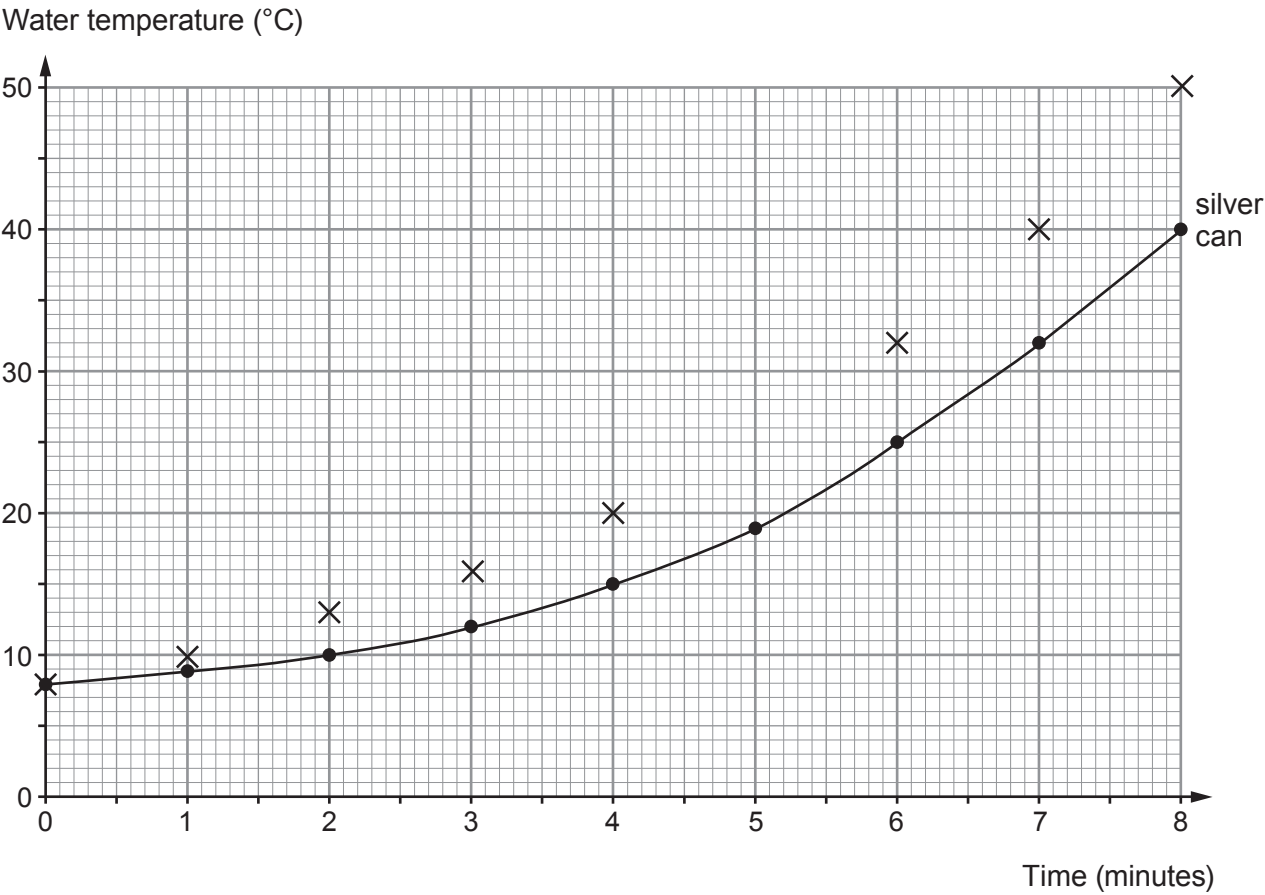
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(b) Each minute, the temperature of the water in each can was measured.

The data, for the silver can, have been plotted on the grid below and its best fit curve drawn.

Apart from one result, the data points for the black can were also plotted on the grid.



(i) At **5 minutes** the water temperature in the black can was **25°C** .

Plot this missing data point on the grid above **and** add the line of best fit. [2]

(ii) Compare the temperature rise in the black can with the temperature rise in the silver can.

Explain your answer in terms of heat radiation. [2]

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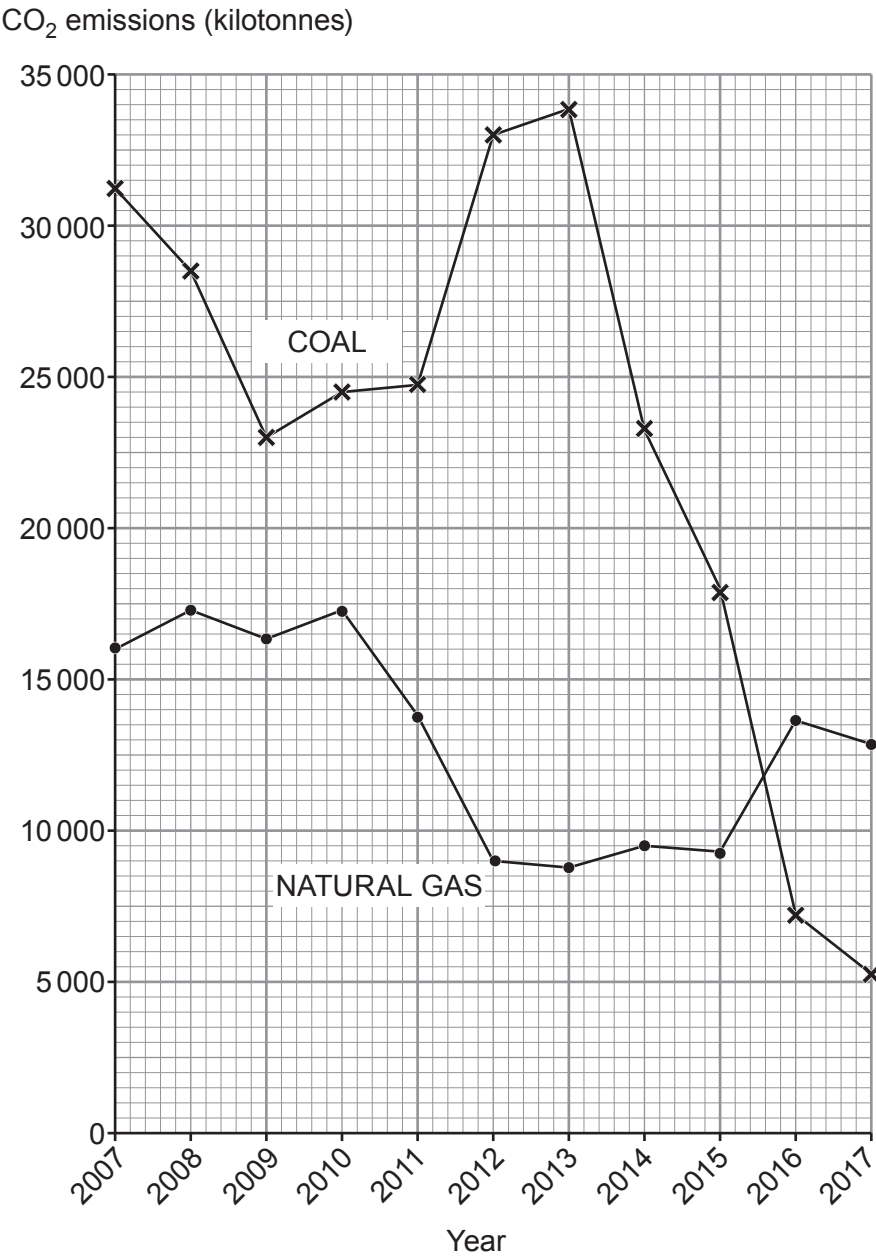
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6. The UK government wants to reduce the CO₂ emissions from power stations.

The graphs below show the CO₂ emissions from coal and natural gas power stations between the years 2007 and 2017.



- (a) Look at the data for **2012**. Calculate the difference in the emissions of CO₂ from coal and natural gas. [2]

difference = kilotonnes of CO₂

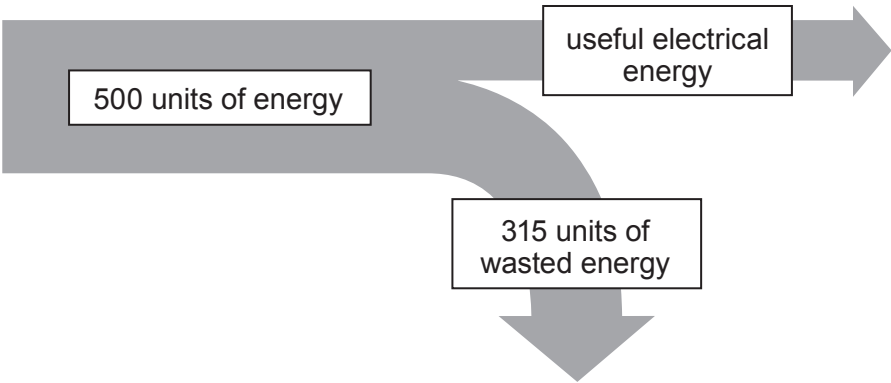
- (b) Between the years 2014 and 2015, the emission of CO₂ from coal fell by 5500 kilotonnes.
- State between which other years the emission of CO₂ from coal fell **at the same rate**. [1]
- Years and

- (c) State **two** benefits of reducing CO₂ in the atmosphere. [2]
1.
2.

- (d) (i) Nuclear power stations provide up to 20% of the present UK demand for electricity.
- Gas provides up to 50%.
- One student, Seren, says that a graph for the CO₂ emissions from nuclear power stations would be the same shape as for gas but always lower.
- Explain whether you agree with Seren. [2]
-
-
-



(ii) The Sankey diagram shows the energy input and output for a power station.



Seren looks at the diagram and calculates that the power station is 63% efficient.

Explain whether you agree with Seren. [2]

Space for calculation.

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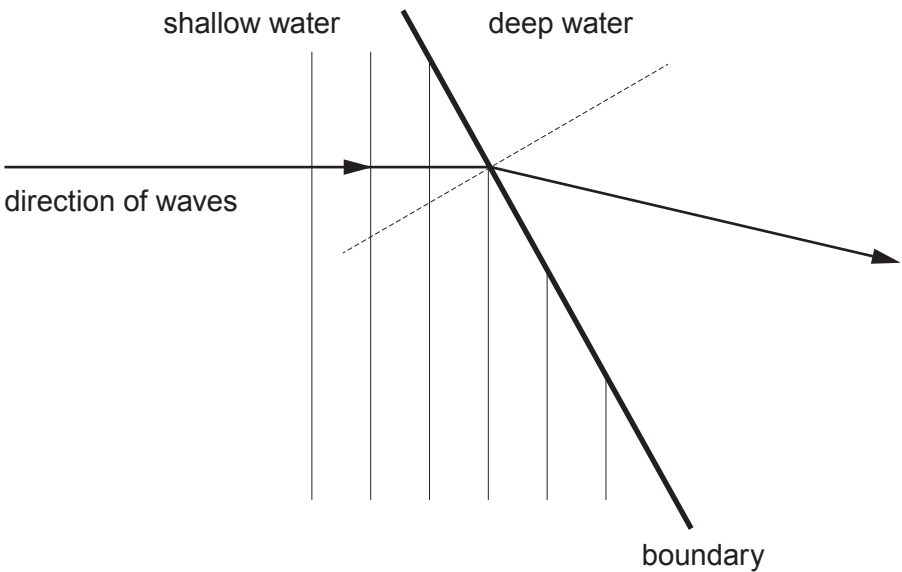


7. Students study refraction of waves in a ripple tank.

They set up the tank to show waves going from shallow water into deeper water.

The diagram shows a number of wavefronts approaching a boundary between shallow and deep water.

The first four wavefronts have reached the boundary.

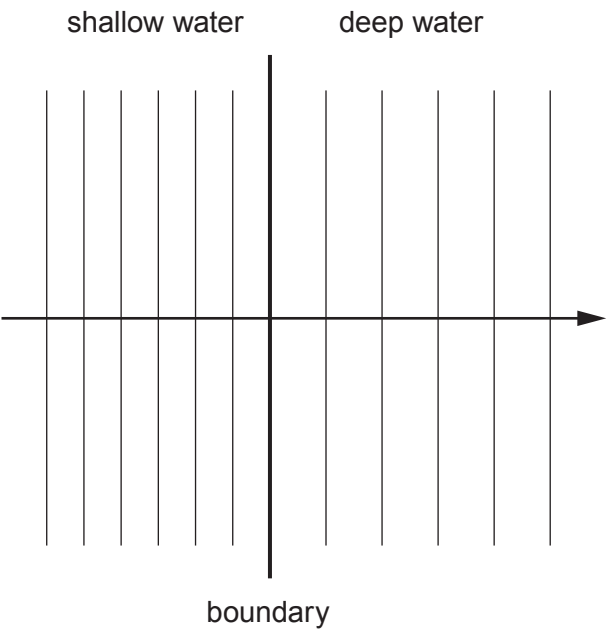


(a) **Complete the diagram** to show the wavefronts in the deep water.

[3]



(b) In a separate experiment, the wavefronts pass over the boundary in the way shown below.



Sam says that the speed of the waves is greater in deep water than shallow water.

Explain whether you agree with his statement. [3]

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END OF PAPER

